INCH-POUND

MIL-A-0052767E(ME)
10 May 1994
USED-IN-LIEU-OF
MIL-A-52767D
8 February 1989

MILITARY SPECIFICATION

AIR CONDITIONERS: VERTICAL AND HORIZONTAL, COMPACT

This specification is approved for use by the US Army Belvoir Research, Development, and Engineering Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-A-52767D. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers compact, self-contained, vertical, and horizontal air conditioners.

1.2 Classification. The air conditioners shall be of the following types, sizes, and classes, as specified (see 6.2 and 6.6):

Type I - Vertical

Size	Class
A - 6,000 Btu/hr	1 - 115 volt (V), single-phase, 50/60 Hertz (Hz). 2 - 208 V, 3-phase, 400 Hz.
B - 9,000 Btu/hr	1 - 115 V, single-phase, 50/60 Hz. 2 - 230 V, single-phase, 50/60 Hz. 3 - 208 V, 3-phase, 50/60 Hz. 4 - 208 V, 3-phase, 400 Hz.
C - 18,000 Btu/hr	1 - 208 V, 3-phase, 50/60 Hz. 2 - 208 V, 3-phase, 400 Hz.
D - 36,000 Btu/hr	1 - 208 V, 3-phase, 50/60 Hz. 2 - 208 V, 3-phase, 400 Hz.
E - 60,000 Btu/hr	1 - 208 V, 3 phase, 50/60 Hz. 2 - 208 V, 3-phase, 400 Hz.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4120

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

Type II - Horizontal

Type	Class
A - 9,000 Btu/hr	1 - 115 V, single-phase, 50/60 Hz. 2 - 230 V, single-phase, 50/60 Hz. 3 - 208 V, 3-phase, 50/60 Hz. 4 - 208 V, 3-phase, 400 Hz.
B - 18,000 Btu/hr	1 - 230 V, single-phase, 50/60 Hz. 2 - 208 V, 3-phase, 50/60 Hz. 3 - 208 V, 3-phase, 400 Hz.
C - 36,000 Btu/hr	1 - 208 V, 3-phase, 50/60 Hz. 2 - 208 V, 3-phase, 400 Hz.
D - 60,000 Btu/hr	1 - 208 V, 3-phase, 50/60 Hz. 2 - 208 V, 3-phase, 400 Hz.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1588	- Sack, Shipping, Paper (Cushioned with Closed Cell Plastic Film).
A-A-1898	- Cushioning Material, Cellulosic Packaging.
BB-F-1421	- Fluorocarbon Refrigerants.
NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-1752	- Cushioning Material, Packaging, Polyethylene Foam.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-152	- Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronics and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurement.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-781	- Reliability Testing for Engineering Development, Qualification, and Production.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1474	- Noise Limits for Military Materiel (Metric).

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

TA13216E6510	- Air Conditioner, Horizontal, Compact, 60,000 Btu/hr, 208 Volt, 3-Phase, 50/60 Hertz.
TA13216E6520	- Air Conditioner, Horizontal, Compact, 60,000 Btu/hr, 208 Volt, 3-Phase, 400 Hertz.
TA13217E6897	- Air Conditioner, Vertical, Compact, 60,000 Btu/Hr, 208 Volt, 3-Phase, 50/60 Hertz.
TA13217E6898	- Air Conditioner, Vertical, Compact, 60,000 Btu/hr, 208 Volt, 3-Phase, 400 Hertz.
TA13218E9880	- Air Conditioner, Vertical, Compact, 6,000 Btu/hr, 208 Volt, 3-Phase, 400 Hertz.
TA13218E9890	- Air Conditioner, Vertical, Compact, 6,000 Btu/Hr, 115 Volt, Single-Phase, 50/60 Hertz.
TA13218E9900	- Air Conditioner, Vertical, Compact, 9,000 Btu/hr, 208 Volt, 3-Phase, 400 Hertz.
TA13218E9905	- Air Conditioner, Vertical, Compact, 9,000 Btu/hr, 115 Volt, Single-Phase, 50/60 Hertz.
TA13218E9910	- Air Conditioner, Vertical, Compact, 9,000 Btu/hr, 208 Volt, 3-Phase, 50/60 Hertz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TA13218E9915 - Air Conditioner, Vertical, Compact, 9,000 Btu/hr, 230 Volt, Single-Phase, 50/60 Hertz.
TA13222E9150 - Air Conditioner, Horizontal, Compact, 18,000 Btu/hr, 230 Volt, Single-Phase, 50/60 Hertz.
TA13222E9160 - Air Conditioner, Horizontal, Compact, 18,000 Btu/hr, 208 Volt, 3-Phase, 50/60 Hertz.
TA13222E9170 - Air Conditioner, Horizontal, Compact, 18,000 Btu/hr, 208 Volt, 3-Phase, 400 Hertz.
TA13225E8000 - Air Conditioner, Vertical, Compact, 18,000 Btu/hr, 208 Volt, 3-Phase, 50/60 Hertz.
TA13225E8010 - Air Conditioner, Vertical, Compact, 18,000 Btu/hr, 208 Volt, 3-Phase, 400 Hertz.
TA13225E8200 - Air Conditioner, Vertical, Compact, 36,000 Btu/hr, 208 Volt, 3-Phase, 50/60 Hertz.
TA13225E8210 - Air Conditioner, Vertical, Compact, 36,000 Btu/hr, 208 Volt, 3-Phase, 400 Hertz.
TA13225E8445 - Air Conditioner, Horizontal, Compact, 9,000 Btu/hr, 230 Volt, Single-Phase, 50/60 Hertz.
TA13225E8450 - Air Conditioner, Horizontal, Compact, 9,000 Btu/hr, 115 Volt, Single-Phase, 50/60 Hertz.
TA13225E8455 - Air Conditioner, Horizontal, Compact, 9,000 Btu/hr, 208 Volt, 3-Phase, 50/60 Hertz.
TA13225E8460 - Air Conditioner, Horizontal, Compact, 9,000 Btu/hr, 208 Volt, 3-Phase, 400 Hertz.
TA13225E9500 - Air Conditioner, Horizontal, Compact, 36,000 Btu/hr, 208 Volt, 3-Phase, 50/60 Hertz.
TA13225E9600 - Air Conditioner, Horizontal, Compact, 36,000 Btu/hr, 208 Volt, 3-Phase, 400 Hertz.
TA13229E6149 - Air Conditioner, Horizontal, Compact, 36,000 Btu/hr, 208 Volt, 3-Phase, 50/60 Hertz.
TA13229E6468 - Air Conditioner, Vertical, Compact, 36,000 Btu/hr, 208 Volt, 3-Phase, 400 Hertz.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: US ARMY BELVOIR RDE CTR, ATTN SATBE FES, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS, INC. (ASHRAE)

STD 16 - Method of Testing for Rating Room Air Conditioners and Packaged Terminal Air Conditioners.
STD 37 - Methods of Testing for Rating Unitary Air Conditioning and Heat Pump Equipment.
STD 51 - Laboratory Methods of Testing Fans for Rating.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to: AMERCN SCTY OF HTG, REFRIG, & AIR COND ENGR, 1791 TULLIE CIR, NE, ATLANTA GA 30329.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to: AMERCN SCTY OF MECHL ENGRS, 345 E 47TH STRET, NEW YORK NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Commercial Packaging.
D 3953 - Strapping, Flat Steel and Seals.

(Application for copies should be addressed to: AMERCN SCTY & MTRLS, 1916 RACE STRET, PHILADELPHIA PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel.
D1.2 - Structural Welding Code - Aluminum.

(Application for copies should be addressed to: AMERCN WELD SCTY, 550 NW LEJEUNE RD, PO BOX 351040, MIAMI FL 33135.)

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT (SCAQMCD)

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to: S MGMT CNTRL DIST, 9150 E FLAIR DR, EL MONTE CA 91731.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The air conditioners shall be in accordance with the following top assemblies and as specified herein:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

Size	Class
	Type I
A	1 - TA13218E9890 2 - TA13218E9880
B	1 - TA13218E9905 2 - TA13218E9915 3 - TA13218E9910 4 - TA13218E9900
C	1 - TA13225E8000 2 - TA13225E8010
D	1 - TA13225E8200 2 - TA13225E8210 2 - TA13229E6468
E	1 - TA13217E6897 2 - TA13217E6898
	Type II
A	1 - TA13225E8450 2 - TA13225E8445 3 - TA13225E8455 4 - TA13225E8460
B	1 - TA13222E9150 2 - TA13222E9160 3 - TA13222E9170
C	1 - TA13225E9500 1 - TA13229E6149 2 - TA13225E9600
D	1 - TA13216E6510 2 - TA13216E6520

A control box shall be mounted integrally with the air conditioner. When specified (see 6.2), a block-off assembly shall be furnished. When specified (see 6.2), air conditioners shall be furnished with external sound attenuators as shown on the drawings. When specified (see 6.2 and 6.5), a control adapter assembly, power adapter, blast damper assembly, fresh-air cover, and a special air conditioner cover shall be provided. When specified (see 6.2), air conditioners shall be furnished with nuclear hardened assembly. The weight of each air conditioner, including the control box, resilient mounts, and a full charge of refrigerant and oil (but not including the sound attenuator and block-off assembly, if furnished) shall be not more than as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE I. Maximum weight.

Air conditioner	Maximum weight (lb)
Type I	
Size A, Class 1	180
Class 2	175
Size B, Class 1	200
Class 2	200
Class 3	190
Class 4	190
Size C, Class 1	280
Class 2	280
Size D, Class 1	460
Class 2	460
Size E, Class 1	620
Class 2	620
Type II	
Size A, Class 1	200
Class 2	200
Class 3	200
Class 4	200
Size B, Class 1	290
Class 2	290
Class 3	290
Size C, Class 1	435
Class 2	435
Size D, Class 1	600
Class 2	600

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the air conditioners. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request, for inspection by the contracting officer or the designated representative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor, and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The air conditioners shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the air conditioners may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the air conditioners may be newly fabricated from recovered materials to the maximum extent practicable, provided the air conditioners produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the air conditioners.

3.4 Refrigeration system.

3.4.1 Refrigeration system type. The refrigeration system shall be of vapor-cycle type employing refrigerant conforming to BB-F-1421, type 22, and shall be furnished with a full charge in the system.

3.4.2 Refrigerant system leak. The entire refrigerant system shall withstand pressures up to 425 psig without permanent deformation of any components or any detectable leaks.

3.4.3 Function. Refrigerant flow to the evaporator shall be controlled by the solenoid-liquid-shutoff valve and the evaporator thermal expansion valve. When the predetermined temperature is attained within the conditioned space, the thermostat shall close the solenoid-liquid-shutoff valve and stop the flow of refrigerant to the evaporator. The fluid-pressure regulator valve in the bypass system shall open and bypass the refrigerant gas to the compressor suction and maintain a compressor inlet pressure of not less than 50 psig. An expansion valve (quench valve) in the bypass system reduces the suction gas temperature to tolerable limits and prevents possible damage.

3.4.4 Moisture content. The moisture content in the system shall be not more than 25 parts per million (ppm) by weight in the liquid phase.

3.5 Electrical controls. The controls shall function as specified in 3.5.1 through 3.5.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.5.1 Mode selector switch. The mode selector switch shall control operation of the air conditioner as follows: When the switch is in the "off" position and with power connected to the air conditioner, the compressor crankcase heater shall be energized, and the fan motor(s), heaters, and compressor shall be de-energized and inoperative. When the switch is in the "ventilate" position, the evaporator fan motor shall start and operate. When the switch is in the "cool" position, the evaporator fan motor shall continue to operate and the condenser fan motor (if a separate motor exists) shall start and operate. After a time delay, the compressor motor shall start and operate. Both banks of electrical heaters shall be de-energized when the air conditioner is operating in either the ventilating or cooling mode. When the switch is in "low heat" position, the evaporator fan motor shall start and operate, and when heat is required, one bank of electrical heaters shall be energized. When the switch is in the "high heat" position, the evaporator fan motor shall continue to operate, and when more heat is required than one bank of heaters can provide, both banks of electrical heaters shall be energized. The compressor and condenser fan motor(s) shall not operate when the air conditioner is operating on heating cycle.

3.5.2 Conditioned-air thermostat. The conditioned-air thermostat shall control the cooling and heating temperatures automatically by regulating the action of the solenoid-liquid-shutoff valve in the cooling mode or one bank of electrical heaters in the heating mode.

3.5.3 Dual-speed evaporator motor switch. The dual-speed motor switch, when applicable, shall control the speed of the evaporator or evaporator-condenser fan motor. With the switch in the "high" position, the motor shall operate at high speed; with the switch in the "low" position, the motor shall operate at low speed.

3.5.4 Pressure control switch. The pressure control switch, when applicable, shall automatically control the speed of the evaporator-condenser fan motor when the dual-speed motor switch is in the "low" position. When the condenser pressure reaches the fan high-speed cut-in pressure specified in table II, the pressure control switch shall override the "low" position of the dual-speed motor switch and automatically switch the evaporator condenser fan motor to high speed; when the condenser pressure reaches the fan low-speed cut-in pressure specified in table II, the pressure control switch shall automatically switch the evaporator-condenser fan motor back to low speed.

3.5.5 Heater high-limit cutout switch. The heater high-limit cutout switch shall de-energize the electrical heating elements to prevent overheating or damage to the air conditioner.

3.5.6 High-pressure cutout switch. The high-pressure cutout switch shall be set to stop the compressor when the condenser pressure reaches the pressure specified in table II.

3.5.7 Low-pressure cutout switch. The low-pressure cutout switch shall be set to stop the compressor when the suction pressure decreases to the pressure specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE II. High and low cutout pressures and fan speed cut-in pressure.

	Compressor high-pressure cutout (±10 psig)	Compressor low-pressure cutout (±5 psig)	Fan high-speed cut-in (psig)	Fan low-speed cut-in (psig)
Type I				
Size				
A	480	25	391 ±31	285 ±17
B	480	25	391 ±31	285 ±17
C	480	7	NA	NA
D	480	7	NA	NA
E	480	7	NA	NA
Type II				
Size				
A	480	15	NA	NA
B	480	15	NA	NA
C	480	15	NA	NA
D	480	15	NA	NA

3.6 Wiring. Wiring shall be secured in compact harnesses and attached to frame members with insulated clamps at close intervals to ensure a neat and orderly cable run. Cables and wires shall not be spliced at any point throughout the length of their runs. Wiring shall not be attached to panels or other components that require removal during maintenance, except for actual electrical connections to components. Electrical isolators shall be used between all wiring and metal components to prevent abrasive action to wire insulation.

3.7 Electromagnetic interference. When tested as specified in 4.5.2.6, the electromagnetic interference emission characteristics of the air conditioner shall not exceed the following:

a. The values given on figures 1 and 2 for steady state conditions.
b. Transients above the curves given on figures 1 and 2 in excess of 1-second duration.

3.8 Evaporator compartment air leakage. The air leakage of the evaporator compartment shall be not more than 20 standard cubic feet per minute (scfm) when the evaporator compartment is subjected to an internal pressure of 1.0-inch wg with the evaporator air discharge, return air, and condensate drains closed, and with a duct attached to the chemical-biological-radiological (CBR) opening for supply air, when applicable. The fresh-air damper shall be closed if both a CBR opening and a fresh-air opening are present. If only a fresh-air opening is present, it shall remain open and the duct shall be attached to it.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.9 Performance characteristics.

3.9.1 Capacity. When operating at the specified conditions, the total cooling capacity and the sensible cooling capacity of the air conditioner shall be not less than as shown in table III.

TABLE III. Cooling capacities.

Unit	Condition	Ambient Temperature (dry bulb)	Evaporator inlet temperature		Minimum net sensible cooling capacity	Minimum net total cooling capacity
			dry bulb	wet bulb		
		°F	°F	°F	Btu/hr	Btu/hr
Type I						
Size A	A	120	90	75	3,780	6,300
	B	95	80	67	4,095	6,300
Size B	A	120	90	75	5,600	9,350
	B	95	80	67	6,000	9,350
Size C	A	120	90	75	12,100	18,500
	B	95	80	67	14,000	21,500
Size D	A	120	90	75	22,680	37,800
	B	95	80	67	24,570	37,800
Size E	A	120	90	75	36,000	60,000
	B	95	80	67	39,000	60,000
Type II						
Size A	A	120	90	75	8,000	10,000
	B	95	80	67	8,000	10,500
Size B	A	120	90	75	14,000	18,500
	B	95	80	67	14,700	19,500
Size C	A	120	90	75	30,000	41,000
	B	95	80	67	32,000	40,000
Size D	A	120	90	75	44,000	62,000
	B	95	80	67	48,000	63,000

Note: Data shown for 0-inch water gauge (wg) (free air) external static pressure on the evaporator side for all units except type II, size B (type II, size B data are shown for 0.25-inch wg external static pressure on the evaporator side).

3.9.2 Evaporator airflow. With the mode selector switch positioned on "ventilate" and with the fan at high speed, the evaporator airflow in standard cubic feet per minute (scfm) shall be not less than as specified in table IV. The airflow shall be obtained with the external static pressure resistance in inch-wg as specified in the note to table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE IV. Evaporator airflow.

Size	Class	Airflow minimum (scfm)
Type I		
A	1	280
	2	300
B	1	335
	2	335
	3	335
	4	335
C	1	585
	2	615
D	1	1250
	2	1360
E	1	1900
	2	2020
Type II		
A	1	370
	2	370
	3	370
	4	370
B	1	590
	2	590
	3	640
C	1	1380
	2	1520
D	1	1800
	2	2000

3.9.3 Power consumption. When the air conditioners are operating at the conditions shown in table III, the total power consumption shall be not more than specified in table V at a power factor not less than specified in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE V. Power consumption and power factor.

Size	Class	Power consumption (maximum) (kW)	Power Factor (minimum)
Type I			
A	1	2.20	0.90
	2	2.60	0.50
B	1	3.60	0.95
	2	3.40	0.95
	3	3.40	0.84
	4	3.60	0.55
C	1	5.00	0.90
	2	5.70	0.61
D	1	9.60	0.83
	2	11.00	0.60
E	1	15.50	0.78
	2	18.60	0.63
Type II			
A	1	3.22	0.85
	2	3.22	0.90
	3	3.00	0.71
	4	3.10	0.45
B	1	5.00	0.95
	2	4.60	0.80
	3	5.00	0.55
C	1	13.50	0.83
	2	13.50	0.64
D	1	14.00	0.85
	2	18.00	0.67

3.9.4 Operating pressures. When the air conditioners are operating as specified in 3.9.1 or 3.9.6, the condensing and suction pressures shall be as specified in table VI.

3.9.5 Heating. With all heaters operating on high heat in an ambient temperature of 75 to 80 °F and the dual-speed motor switch, when applicable, positioned on "high", the heating capacity shall be not less than as specified in table VII. Electrical potential from the air-conditioner casing to the ground shall be not more than 1 volt (V). Power consumption shall be not more than as specified in table V.

3.9.6 High temperature operation. After being subjected to not less than 120 °F for a period of 4 hours, the air conditioner shall start and operate in the cooling mode in an ambient temperature of not less than 120 °F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE VI. Condensing and suction pressure.

Size	Class	Pressures (psig)				
		Operating conditions (see 3.9.1)		Operating conditions (see 3.9.6)		
		Condensing pressure	Suction pressure	Condensing pressure		Suction pressure
		(max)	(min)	(max)	(min)	(min)
Type I						
A	1	405	65	470	N/A	50
	2	405	65	470	N/A	50
B	1	405	65	470	N/A	50
	2	405	65	470	N/A	50
	3	405	65	470	N/A	50
	4	405	65	470	N/A	50
C	1	440	65	470	N/A	50
	2	440	65	470	N/A	50
D	1	430	65	470	105	50
	2	430	65	470	105	50
E	1	420	65	470	105	50
	2	420	65	470	105	50
Type II						
A	1	425	50	470	160	50
	2	425	50	470	160	50
	3	425	50	470	160	50
	4	425	50	470	160	50
B	1	425	50	470	160	50
	2	425	50	470	160	50
	3	425	50	470	160	50
C	1	430	50	470	160	50
	2	430	50	470	160	50
D	1	430	50	470	160	50
	2	430	50	470	160	50

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE VII. Heating capacity.

Size	Class	Heating capacity minimum (Btu/hr)
Type I		
A	1	4,500
	2	4,500
B	1	6,000
	2	6,000
	3	6,000
	4	6,000
C	1	12,000
	2	12,000
D	1	28,600
	2	28,600
E	1	47,000
	2	47,000
Type II		
A	1	7,000
	2	7,000
	3	7,000
	4	7,000
B	1	13,400
	2	13,400
	3	13,400
C	1	31,200
	2	34,700
D	1	45,000
	2	47,000

3.10 Rain. When subjected to the tests specified in 4.5.2.11, the air conditioners shall operate without functional damage or impairment of operation. There shall be no penetration of water from the evaporator section to the dry side of the enclosure or into or on any electrical components; in particular, the motor, electrical connector, terminal switch, or control that could result in an electrical short, cause damage from rust or corrosion, or become a hazard to personnel.

3.11 Salt fog. When subjected to the tests specified in 4.5.2.12, no part of the air conditioner shall show evidence of deterioration, corrosion, or change in tolerance limits that could prevent the unit from meeting operational requirements. The unit shall show no evidence of peeling of paint, malfunctioning, refrigerant leaks, rough operation, electrical shorts, erratic pressure, or other irregular operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.12 Vibration. When subjected to the test specified in 4.5.2.13, the air conditioners shall show no evidence of structural damage, misalignment or malfunction of components, leaks, fractures, abnormal vibration, or any other irregular operation.

3.13 Reliability. When the air conditioner is operated at rated capacity and tested as specified in 4.5.2.15, the mean-time-between-failures (MTBF) shall be as follows:

Upper Test MTBF (θ_0) = 960 hrs
Lower Test MTBF (θ_1) = 480 hrs

3.14 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be treated for the effects of fungus growth and moisture as specified in MIL-T-152 with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.14.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces. Surfaces shall be previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.14.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 442, South Coast Air Quality Management Control District.

3.15 Treatment and painting. All parts shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable, with the following exceptions:

a. Internal components previously painted by the manufacturer need not be repainted when they conform to the salt/fog requirements.
b. Coils, refrigerant tubing, electrical components, or other components whose operation will be impaired by painting shall not be painted.
c. Unless otherwise specified (see 6.2), the top coat shall be camouflage green 383 in accordance with FED-STD-595.

3.16 Transportability. The air conditioners shall be capable of being transported without damage by military van or truck on cross-country roads at variable speeds up to 20 mph when mounted as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

a. Type I, sizes A, B, and C air conditioners: The air conditioners shall be mounted by bolts using the mounting receptacles provided in the bottom of and above the condenser-fan discharge grille with mounting fixtures as shown on figure 3.
b. Type I, sizes D and E air conditioners: The air conditioners shall be mounted by bolts using the mounting receptacles provided in the bottom of the air conditioner, and by the upper restraining channel and bars shown on figure 3.
c. Type II, sizes A, B, and C air conditioners: The air conditioners with the resilient mounts specified in the drawings shall be mounted by bolts using the mounting receptacles provided in the bottom of the air conditioner.
d. Type II, size D air conditioners: The air conditioners shall be mounted by bolts using the mounting receptacles provided in the bottom of the air conditioner.

3.17 Maintainability.

3.17.1 Maintenance ratio. The air conditioners shall have a maintenance ratio of not more than 0.03. The maintenance ratio is defined as the ratio of the total active maintenance manhours required (scheduled and unscheduled) to the total operating time. Manhours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing. Not more than 25 percent of repairs shall require maintenance at the general support level.

3.17.2 Scheduled maintenance. The air conditioners shall not require scheduled maintenance more often than every 250 hrs of operation. Scheduled maintenance shall not be required at any level higher than organizational maintenance.

3.18 Safety. All rotating or reciprocating parts and other parts subject to high operational temperatures that are of such a nature or are so located as to be a hazard to operating personnel shall be guarded or insulated to the extent necessary to eliminate the hazard. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.

3.19 Blast damper. When specified (see 6.2 and 6.5), the blast damper shall permit full fresh-air flow when in the open position and shall remain open at 2.5 inches water gauge (wg). The damper shall close and remain closed at 3.6 wg. The damper shall withstand a pressure of not less than 3.4 psig. The damper shall not leak, rupture, or be permanently deformed as a result of the 3.4-psig blast.

3.20 Noise limits.

3.20.1 Limit category. When the type and size air conditioner is tested in accordance with 4.5.2.20, the noise limit listed in table VIII shall not be exceeded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE VIII. Noise limit category.

Air conditioner		Noise limit dB (0.0002 microbar)	
Type	Size	Evaporator	Condenser
I	A	MIL-STD-1474, table II, E	MIL-STD-1474, table II, D
I	B	MIL-STD-1474, table II, E	MIL-STD-1474, table II, D
I	C	MIL-A-52767, table IX	MIL-A-52767, table IX
I	D	MIL-STD-1474, table II, D	MIL-STD-1474, table II, E + 5
I	E	MIL-STD-1474, table II, D	MIL-STD-1474, table II, E + 5
II	A	MIL-A-52767, table IX	MIL-A-52767, table IX
II	B	MIL-A-52767, table IX	MIL-A-52767, table IX
II	C	MIL-A-52767, table IX	MIL-A-52767, table IX
II	D	MIL-STD-1474, table II, D	MIL-STD-1474, table II, D

3.20.2 Noise hazard caution sign. When tested per 4.5.2.20, a noise hazard caution sign shall be posted on air conditioners that exceeds 85dB(A) at a distance of 48 inches or greater from the evaporator or condenser side. The sign will be per MIL-STD-1474, but modified to indicate the maximum distance from the air conditioner that a noise level of 85dB(A) is obtained. For example, if the noise level of 85dB(A) is obtained at a maximum distance of 9 feet from the condenser side of the air conditioner, the sign, per MIL-STD-1474, will be modified to read, "CAUTION, HIGH INTENSITY NOISE, HEARING PROTECTION REQUIRED WITHIN 9 FEET", and will be posted on the condenser side of the air conditioner.

3.21 Workmanship.

3.21.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. All bends shall be made by controlled means to ensure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE IX. Corrected sound pressure levels in decibels.

Octave band center Frequencies - Hz	Evaporator side (dB max)	Condenser side (dB max)
TYPE I Size C		
63	65	60
125	76	73
250	75	74
500	77	71
1000	69	71
2000	66	70
4000	62	69
8000	56	69
TYPE II Size A		
63	55	70
125	76	76
250	64	63
500	65	64
1000	65	63
2000	60	60
4000	53	57
8000	48	49
Size B		
63	60	60
125	66	64
250	69	63
500	68	65
1000	65	66
2000	60	66
4000	64	64
8000	65	55
Size C		
63	71	69
125	70	68
250	70	72
500	69	73
1000	76	70
2000	64	65
4000	63	63
8000	53	51

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

3.21.2 Bolted connections. Bolted connections shall be as specified in the applicable drawings.

3.21.3 Riveted connections. Riveted connections shall be as specified in the applicable drawings.

3.21.4 Welders and welding.

3.21.4.1 Welders and welding operators. Welder and welding operators assigned to manual welding work covered by this specification shall be qualified in accordance with AWS D1.1, AWS D1.2, or the ASME Boiler and Pressure Vessel code for the materials joined and the type of welding operation to be performed. The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes (see 6.7).

3.21.4.2 Welding. The surfaces of parts to be welded shall be free from dirt, scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc (SMA) or gas metal-arc (GMA) process. All welded parts shall be free from slag, flux, weld spatter, cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wire brushed clean.

3.21.4.2.1 Steel welding. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed.

3.21.4.2.2 Aluminum welding. Aluminum members to be welded may be preheated for welding to a temperature of not more than 400 °F. Aluminum welds may be filed, wire brushed, or chipped but shall not be hammered.

3.21.5 Brazing. During the brazing operation a small amount of nonoxidizing gas shall be continuously bled through the refrigerant lines to prevent internal oxide scaling. During the brazing operation, care shall be taken to protect all components from deformation and damage. The quantity of brazing alloy and flux used shall be no more than that required to make the joint. Flux shall not be present in the refrigerant system and heavy beads and fillets of braze material shall be avoided.

3.21.6 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, and unless disapproved by the Government, the contractor's own or any other facilities suitable for the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Disassembly inspection. Failure of any test or examination of the first article shall be cause for disassembly of the first article, in the presence of a Government representative, to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and drawings referenced with regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.1.3 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection. The first article inspection shall be performed on five complete air conditioner assemblies (see 3.2 and 6.3).

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection and cause for performing the inspection specified in 4.1.2 as applicable.

4.3.2 Tests. The first article shall be subjected to the test marked "X" in column 2 through 6 of table X in the sequence listed. Failure of any test shall be cause for rejection and performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.4.2 Examination. Sample selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be subjected to the test marked "X" in column 7 of table X in the sequence listed. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The air conditioner shall be examined as specified herein for the following defects:

101. Weight of air conditioners not as specified (see 3.1).
102. Parts or components missing (see 3.1).
103. Washers or lockwashers missing (see 3.1).
104. Nonconformance to the drawings (see 3.1.1).
105. Ventilation damper does not open fully or close tightly, or is not adjustable (see 3.1).
106. Materials not as specified (see 3.3).
107. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
108. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
109. Used, rebuilt or remanufactured components, pieces or parts incorporated in the air conditioner (see 3.3.2).
110. Air conditioners not fully charged with the type of refrigerant specified (see 3.4.1).
111. Installation of wiring not as specified (see 3.6).
112. Electromagnetic interference suppression components and network not identical to approved model (see 3.7).
113. Electrical circuitry not protected from the effects of moisture and fungus growth (see 3.14).
114. Treatment and painting not as specified or painting defects such as blisters, thin spots, lumps, scratches, runs or sags (see 3.15).
115. Color not as specified (see 3.15).
116. Metal fabrication not as specified (see 3.21.1).
117. Bolted connections not as specified (see 3.21.2).
118. Riveted connections not as specified (see 3.21.3).
119. Welders and welding operators not qualified as specified (see 3.21.4.1).
120. Welding not as specified (see 3.21.4.2).
121. Brazing not as specified (see 3.21.5).
122. Interchangeability not as specified (see 3.21.6).

4.5.2 Tests

4.5.2.1 Test schedule.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE X. Test schedule.1/

Test req	Type of test						Test	Test paragraph	Requirement paragraph
	First article					Quality Conform			
	#1	#2	#3	#4	#5				
1	2	3	4	5	6	7	8	9	10
1	X	X	X	X	X	X	Refrigeration system pressure.	4.5.2.3.1	3.4.2
2	X	X	X	X	X	X	Examination.	4.5.1	3.1, 3.3, 3.4.1, 3.6, 3.14, 3.15, 3.18, 3.21
3	X	X	X	X	X	X	Controls.	4.5.2.4	3.4.3, 3.5
4	X	-	-	-	-	-	Refrigerant moisture content.	4.5.2.5	3.4.4
5	X	-	-	-	-	-	Electromagnetic interference.	4.5.2.6	3.7
6	X	X	X	X	X	X	Evaporator compartment air leakage.	4.5.2.7	3.8
7	X	-	-	-	-	-	Capacity and airflow.	4.5.2.8	3.9.1, 3.9.2 3.9.3, 3.9.4
8	X	-	-	-	-	-	Heating.	4.5.2.9	3.9.3, 3.9.5
9	-	X	-	-	-	-	High temperature.	4.5.2.10	3.9.6, 3.9.4
10	-	X	-	-	-	-	Rain.	4.5.2.11	3.10
11	-	X	-	-	-	-	Salt fog.	4.5.2.12	3.11
12	-	-	-	-	-	-	Vibration.	4.5.2.13	3.12
13	X	X	-	-	-	X	Run-in.	4.5.2.14	3.9
14	X	X	X	X	X	X	Refrigeration system leak.	4.5.2.3.2	3.4.2
15	-	-	X	X	X	-	Reliability.	4.5.2.15	3.13
16	X	-	-	-	-	-	Transportability.	4.5.2.16	3.16
17	-	-	X	X	X	-	Maintainability.	4.5.2.17	3.17
18	X	-	-	-	-	-	Safety.	4.5.2.18	3.18
19	X	-	-	-	-	-	Blast damper.	4.5.2.19	3.19
20 2/	X	-	-	-	-	-	Noise limits.	4.5.2.20	3.20

1/ Perform inspection marked "X".
2/ Only when blast damper is required.

4.5.2.2 Conditions. Unless otherwise specified herein, tests shall be conducted at rated voltage and employ 60-Hz or 400-Hz electrical power, as applicable. The frequency tolerance for all tests shall be ±1 percent. Unless otherwise specified herein, no braces, sealing materials, tiedown devices, or other items that are not inherently a part of the operating air conditioner shall be employed. Unless otherwise specified herein, no alignment of components, addition of refrigerant, or other servicing shall be performed on the production model or production air conditioners after testing of the air conditioner has begun. Components subject to atmospheric corrosion shall not be treated with any treatment such as grease or oil to meet test requirements. The external sound

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

attenuator shall not be installed during tests. Unless otherwise specified herein, all tests shall be conducted with the fans operating at high speed. Grilles, filters, and mist eliminators shall be in place.

4.5.2.3 Refrigerant system pressure and leakage.

4.5.2.3.1 Refrigerant system pressure. Charge the refrigerant system to 425 psi, ±5 psig with dry nitrogen containing refrigerant gas detectable with a halogen-sensitive electronic leak detector (General Electric type H-2 or equal). Reduce the pressure to 300, ±5 psig. Calibrate the leak detector for the type of gas and percentage of mixture employed and set the detector to indicate a leakage rate of 0.1 ounce per year corrected to the percentage of the mixture. Check the entire refrigerant system for leaks. Any detectable refrigerant leak or any permanent deformation of any component in the refrigerant system shall constitute failure of this test.

4.5.2.3.2 Refrigerant system leakage. Test for leakage at room ambient temperature (70 °F or above) with a halogen-sensitive electronic leak detector (General Electric type H-2, or equal). Calibrate the leak detector for the type of gas and set the detector to indicate a leakage rate of 0.1 ounce per year. Check the entire refrigerant system for leaks. A detectable leak shall constitute failure of this test.

4.5.2.4 Controls. Electrical controls shall be tested as specified herein. With power connected to the air conditioner, position the mode selector switch and thermostat in the following position:

a. "Off" position. Determine that the air conditioner components have been de-energized as specified in 3.5.1.
b. "Ventilate" position. Determine that the evaporator fan motor is energized and operating, and that the compressor motor, condenser fan motor (if a separate condenser fan motor is furnished), and the heaters are off.
c. "Cool" position-thermostat coolest position. Immerse the conditioned-air thermostat bulb in water or air having a temperature of 70 +15 or -5 °F. Determine that the evaporator and condenser fan motor(s) is energized and operating and the heaters are off. After a time delay, determine that the compressor motor has started and is operating. Determine that there is refrigerant flow through the solenoid liquid shutoff valve to the expansion valve and not through the fluid pressure regulating valve.
d. "Cool" position-thermostat warmest position. Immerse the conditioned-air thermostat bulb in water or air having a temperature not exceeding 80 °F. Determine that the compressor motor and evaporator and condenser fan motor(s) are energized and operating. Determine that there is no refrigerant flow through the solenoid liquid shutoff valve to the expansion valve and that the fluid pressure regulating valve maintains suction pressures of not less than 35 psig.
e. "High heat" position-thermostat warmest position. Immerse the conditioned-air thermostat bulb in water or air having a temperature of 70 -5 or +15 °F. Determine that the evaporator fan motor is energized and operating and the compressor motor and condenser fan motor (if furnished) are off. Determine that both banks of heaters are energized and operating.
f. "High heat" position-thermostat coolest position. Determine that one bank of heaters is off and the other bank of heaters is energized and operating.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

g. "Low heat" position-thermostat warmest position. With the conditioned-air thermostat bulb still immersed in water or air at a temperature of 70 -5 or +15 °F determine that one bank of heaters is energized and operating and the other bank of heaters is off.
h. "Low heat" position-thermostat coolest position. Determine that both banks of heaters are off.

Nonconformance to 3.4.3, 3.5.1, or 3.5.2 shall constitute failure of this test.

4.5.2.4.1 Dual-speed evaporator motor switch. Turn the selector switch to the "cool" position and the dual-speed switch to the "high" position. Determine that the evaporator fan motor is operating at high speed. With the selector switch still in the "cool" position, turn the dual-speed motor switch to the "low" position. Determine that the evaporator fan motor is operating at low speed. Nonconformance to 3.5.3 shall constitute failure of this test.

4.5.2.4.2 Pressure control switch. For type I, size A, and B units only, operate the air conditioners in the cooling mode with the conditioned-air thermostat set for minimum temperature and with the dual-speed motor switch in the "low" position. Cover the condenser coil-intake air to cause the head pressures to increase to the fan high-speed cut-in pressure in table II. Determine that the dual-speed motor switches to high speed. Remove the condenser coil air inlet cover and determine that the dual-speed motor returns to low speed at the fan low-speed cut-in pressure in table II. Nonconformance to 3.5.4 shall constitute failure of this test.

4.5.2.4.3 Heater high-limit cutout switch. Operate the air conditioner on high heating mode. After 15 minutes of operation, de-energize the power to the evaporator fan, thus causing an increase in temperature within the evaporator compartment. Nonconformance to 3.5.5 shall constitute failure of this test.

4.5.2.4.4 High-pressure cutout switch. The high pressure cutout switch shall be tested as follows:

a. First article. With the air conditioner operating in the cooling mode, gradually block off the condenser airflow until the compressor is stopped by action of the high-pressure cutout switch. Record the pressure at which the compressor is stopped. Cutout pressure not within those specified in table II shall constitute failure of this test.
b. Quality conformance. The high-pressure cutout switch shall be tested in the air conditioner with the air conditioner operating in the cooling mode. With the compressor operating, press the high pressure switch reset button. The compressor shall cease to operate. Failure of the compressor to cease operating at the time the reset button is pressed shall constitute failure of this test.

4.5.2.4.5 Low-pressure cutout switch. The low-pressure cutout switch shall be tested as follows:

a. First article. Electrically connect a continuity meter to the pins of the low pressure switch connector, or to the terminals or across the appropriate terminals in the junction box which are equivalent to the low pressure switch terminals. The meter shall show a closed circuit. Connect a low pressure gauge to the suction service valve. Block off the evaporator

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

airflow while constantly monitoring the suction pressure and continuity meter. The continuity meter shall show an open circuit when the suction pressure reduces to the low pressure specified in table II. Cutout pressures not in accordance with those specified in table II shall constitute failure of this test.

b. Quality conformance. The low-pressure cutout switch shall be tested in the air conditioner with the air conditioner operating in the cooling mode. With the compressor operating, press the low pressure switch reset button. The compressor shall cease to operate. Failure of the compressor to cease operating at the time the reset button is pressed shall constitute failure of this test.

4.5.2.5 Refrigerant moisture content. After not less than 1 hour of continuous operation, the cooling cycle in an ambient temperature of not less than 95 °F, shut down the air conditioner. After an additional period of 1 hour, withdraw a sample of approximately 1 pound of refrigerant from the low-pressure side and determine the moisture content by one of the methods specified in BB-F-1421 or by the titrimetic method using the Karl Fischer reagent. A moisture content of more than 25 ppm shall constitute failure of this test.

4.5.2.6 Electromagnetic interference. Using the test plan specified in MIL-STD-462, the air conditioner shall be tested in accordance with MIL-STD-461 using test methods RE-02 (0.15 MHz to 1,000 MHz) and CE-03 (1.5 MHz to 65 MHz) as follows:

a. Steady-state conditions. Controls of the air conditioner shall be set for the following modes of operation at steady-state conditions:

STEADY-STATE TEST MODES

Rotary switch setting	Thermostat setting
Cool	Coolest
Cool	Warmest
High-heat	Warmest (2 heater banks)
High-heat	Coolest (1 heater bank)

b. Transient conditions. Transient durations shall be measured when the air conditioner controls are switched between the following modes:

Mode	Rotary switch setting	Thermostat position
Cool to bypass	Cool	Coolest to warmest
Bypass to cool	Cool	Warmest to coolest
Maximum heat to minimum heat (Two heater banks to one heater bank)	High heat	Warmest to coolest
Minimum heat to maximum heat (One heater bank to two heater banks)	High heat	Coolest to warmest

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.5.2.6.1 Basis of failure. One or more of the following shall constitute failure of the electromagnetic interference test:

a. Steady-state emissions in excess of those shown on figures 1 and 2.
b. Transient emissions above the limits specified on figures 1 and 2 in excess of 1-sec duration.

4.5.2.7 Evaporator compartment air leakage. Close the fresh-air damper if necessary (see 3.8). Box the entire front face (evaporator side) of the air conditioner so that the evaporator is at least 2 inches from the box surface facing it and air is free to circulate over the entire evaporator side (from the top of the evaporator discharge opening to the base of the air conditioner). Seal the box to the air conditioner so that air conditioner and the box does not leak between the air conditioner and the box. Attach a duct equipped with an air-circulating and measuring device to the CBR or fresh-air opening (see 3.8) in the air conditioner. Seal the connection between the air conditioner and duct to prevent external air leakage. Seal condensate drains. Other sealing that is not a part of the air conditioners shall not be permitted except as required for instrumentation lines. By means of the air-circulating device, subject the evaporator compartment to static pressure of 1.0-inch wg and maintain the pressure for not less than 5 min while measuring the airflow leakage. Airflow leakage shall be measured using the instruments, apparatus, and procedures specified in the ASHRAE Standard 37 or 51 (nozzles only). Nonconformance to 3.8 shall constitute failure of this test.

4.5.2.8 Cooling capacity and airflow. Test the air conditioner for cooling capacity at the capacity rating conditions specified in 3.9.1. Test the evaporator fan for airflow at the conditions specified in 3.9.2. Insulating or sealing material that is not inherently a part of the air conditioner shall not be employed.

4.5.2.8.1 Cooling capacity. Capacity tests shall be conducted using the instruments, apparatus, and procedures specified in ASHRAE Standards, as specified in table XI and as specified herein. Tests conducted in accordance with Standard 37 shall use air-enthalpy method for condenser confirmatory test. Capacity tests shall be conducted with type I air conditioners located with the front face (evaporator side) flush with the inside room surface of the wall between the inside and outside room. Type II air conditioners shall be located, with respect to the inside and outside room for the capacity tests, as shown on figure 4. Prior to the capacity tests, thermocouples shall be installed at the thermal expansion valve bulb, quench valve bulb, quench valve discharge line, bypass valve discharge line, liquid line at the expansion valve inlet, and hot gas compressor discharge line. After installation, the thermocouples shall be completely insulated to prevent ambient temperatures from affecting the temperatures obtained from them. In addition to the data required by the applicable ASHRAE standards, temperatures obtained from the above thermocouples shall be recorded at the same times that the ASHRAE data are recorded. These temperatures shall be used, in the event of capacity test failures, to troubleshoot the refrigeration system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE XI. Applicable ASHRAE standards.

Size	Standard
Type I	
A	16
B	16
C	37
D	37
E	37
Type II	
A	16
B	16
C	37
D	37

4.5.2.8.2 Airflow. Airflow tests shall be conducted using the instruments, apparatus, and procedures specified in ASHRAE Standard 37 or ASHRAE Standard 51 (nozzles only for air flow measurement).

4.5.2.8.3 Basis of failure. Nonconformance to 3.9.1, 3.9.2, 3.9.3, 3.9.4, or 4.5.2.2 shall constitute failure of this test.

4.5.2.9 Heating. The test shall be conducted in any ambient temperature of 75 +5 -0 °F. Prior to recording any data, operate the air conditioner for a period of not less than 2 hours with the mode selector switch turned to the "high heat" position and the conditioned-air thermostat set for maximum temperature. Calculate the heating capacity using the following equation:

$$QH = 3.414 \times W$$

where QH = Heating capacity (Btu/hr)
W = Power input (Watts)

Measure the power input (W) to the air conditioner and the electrical leakage (V) from the air-conditioner casing to ground. Nonconformance to 3.9.3 or 3.9.5 shall constitute failure of this test.

4.5.2.9.1 Heating (50-Hz electrical power). After completion of the above test, and when 50/60-Hz air conditioners are specified (see 1.2), repeat the heating test using 50-Hz electrical power. Any evidence of heater malfunction when operating at 50-Hz electrical power shall constitute failure of this test.

4.5.2.10 High-temperature operation cooling. Subject the entire air conditioner to an ambient temperature of not less than 120 °F for a period of 4 hours. After completion of the exposure period and prior to cooling mode operation, nowhere shall air temperature be less than 120 °F. Operate the air conditioner in the cooling mode at the coolest temperature setting, for a period of not less than 1 hour in an ambient temperature of not less than 120 °F. No divider, duct, guide, nor any similar item shall be used on the air conditioner in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

order to influence the direction or the temperature of inlet or outlet air of the air conditioner. When 50/60-Hz air conditioners are specified (see 1.2), repeat the entire test at 50-Hz electrical power. Inability of the air conditioner to start and operate or nonconformance to 3.9.4 shall constitute failure of this test.

4.5.2.11 Rain. The air conditioner shall be installed on an enclosure to simulate a shelter application. The type I air conditioner evaporator section shall be flush with the outside opening of the enclosure. Type II, sizes A, C, and D air conditioner evaporator sections shall protrude through the opening of the enclosure with the outside opening over the center cover. Type II, size B air conditioner evaporator section shall protrude through the opening 4 inches. Seal the opening to prevent any water from entering the enclosure. The condenser covers shall be rolled up and the fresh air doors closed. The test facility, equipment and instrumentation shall be in accordance with MIL-STD-810 method 506, rain. The depth of rainfall shall be measured near the surface of the test unit by a U.S. Weather Bureau type gauge. Water supply shall contain fluorescent type dye for ease in determining the location and extent of leakage. Conduct tests specified in 4.5.2.11.1 and 4.5.2.11.2. The pretest and post-test for each rain test shall consist of visual examination, measurement of the electrical resistance of each insulated electrical conductor segment for conformance to 3.10, and 20 minutes operation in the cooling mode and 20 minutes operation in the heating mode. During this operation, 4.5.2.4 test shall be performed. After each rain test, remove all panels and access doors for visual examination.

4.5.2.11.1 Free rain. Subject each of the three exposed sides tilted at a 45-degree angle, sides up, to a simulated rain fall of 4 inches per hour ±1 inch for 30 minutes. Do not operate air conditioner during test. The total test duration shall be 2 hours.

4.5.2.11.2 Blowing rain. With the air conditioners operating in the cooling mode, subject each of the three exposed sides to a simulated rainfall of 1 inch per hour ±0.50 inch combined with a horizontal wind of 40 ±5 mph for a period of 1 hour. Total test duration shall be not less than 3 hours.

4.5.2.11.3 Basis of failure. One or more of the following shall constitute failure of the rain test:

a. Evidence of water leaking into the enclosure.
b. Malfunctioning of components or an electrical short during operation.
c. Nonconformance to 3.10.

4.5.2.12 Salt fog. Subject the air conditioner to MIL-STD-810, method 509, procedures I. Do not operate the air conditioner during exposure. Prior to this test and after the 48-hour drying period, visually examine the unit, measure the insulation resistance, and conduct the 4.5.2.4 test. Nonconformance to 3.11 and 4.5.2.4 shall constitute failure of this test.

4.5.2.13 Vibration. Conduct the vibration test as specified in 4.5.2.13.1 to the following conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

a. Category: Test item shall be categorized as equipment installed in ground vehicles.
b. Apparatus: Any vibration - inducing equipment with required instrumentation capable of satisfying the test conditions is acceptable.
c. Applicable test: Sinusoidal cycling (see 4.5.2.13.1).
d. Test time schedule (per axis):

(1) Sinusoidal cycling time - 30 minutes (min) per 1000 miles.
(2) Maximum cycling time - 60 min, total.
(3) Sweep time - 12 min, range 5-200-5 hertz (Hz). Sweep time shall be increased by 3 min if test frequency goes to 2 Hz.

e. Test level curve: See figure 5.
f. The air conditioner shall not be operated during this test and shall be mounted as follows:

(1) Type I, sizes A, B, and C, air conditioners: The air conditioners shall be mounted by bolts using the mounting receptacles provided in the bottom and above the condenser-fan discharge grille.
(2) Type I, sizes D and E air conditioners: The air conditioner shall be mounted using a test fixture as shown in figure 3.
(3) Type II air conditioners: The air conditioners shall be mounted by bolts using the mounting plates provided in the bottom.
(4) Auxiliary braces, straps, or other tiedown devices, other than those specified herein, shall not be employed to secure the air conditioners to the test apparatus. No blocking or tiedown of internal components other than those that are part of the air conditioner shall be employed. Removable panels and access doors shall be in place during all testing. With the air conditioner mounted, there shall be no resonance inherent in the test apparatus at the frequencies and acceleration levels specified for this test. During the tests, the base of the air conditioner shall remain in a horizontal plane. With the exception of type I, sizes D and E air conditioners, mount accelerometers on the test apparatus immediately adjacent to the four corners of the base of the air conditioner. For type I, sizes D and E air conditioners, four accelerometers shall be mounted on the test fixture as shown on figure 3. The four accelerometers shall be connected into an electric or electronic network which will average the accelerometers' output signals into a single indicated g level of the magnitude specified for this test. After completion of each test axis, conduct the controls test specified in 4.5.2.4 and the refrigeration system leak test specified in 4.5.2.3.2. Nonconformance to 3.12, 4.5.2.3.2, or 4.5.2.4 shall constitute failure of this test.

4.5.2.13.1 Sinusoidal vibration test. The vibration shall be applied along each of the three mutually perpendicular axes of the test item in accordance with the test level, frequency range, and times specified in 4.5.2.13. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 5. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending sweep time shown on figure 5 for the specified range. The vibratory accelerator level or double amplitude of the test curve shall be maintained at the test item mounting points.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.5.2.14 Run-in.

4.5.2.14.1 Run-in performance. Install ducts on the evaporator-return-air openings, evaporator-air-discharge opening, and condenser air inlet opening of each air conditioner. Each duct shall be not less than 12 inches long and of the same cross-sectional dimensions as the respective openings. Equip each duct with a thermocouple grid to measure average entering and leaving air temperatures to the evaporator and entering air to the condenser. Measure the entering evaporator-air wet-bulb temperature by psychrometer device located in the test chamber. Operate the air conditioners for not less than 1 hour in the cooling mode while maintaining the supply voltage to the air conditioner within ±5 percent. The air temperature entering the evaporator return and condenser inlet shall be as specified in table XII. After stabilization, three sets of the data specified in table XIII shall be recorded and averaged during the last 30 minutes of operation (every 10 minutes). After completion of the above, operate the air conditioner in an ambient temperature between 100 and 120 °F for a total period of 48 hours in the cooling mode, in cyclic operation of 55 minutes "on" and 5 min "off". After completion of the 48-hour operation, continue operating the air conditioner in the cooling mode for not less than an additional hour while maintaining the supply voltage to the air conditioner within ±5 percent and temperature of the air entering the evaporator return and condenser inlet as specified in table XII. After stabilization, three sets of the data specified in table XIII shall be recorded and averaged during the last 30 minutes of operation (every 10 minutes). With the air conditioner at room ambient temperature, operate the air conditioner on high heat mode for 30 minutes. During the last 10 minutes of operation, record the power input (W) to the air conditioner. After completion of the 30-minute operating period, conduct the refrigerant system leak test as specified in 4.5.2.3.2.

TABLE XII. Temperature conditions for run-in performance.

Location	Temperature
a. Evaporator return.	Dry bulb - 90 ±3 °F.
b. Evaporator return.	Wet bulb - dry-coil condition.
c. Condenser inlet.	Dry bulb - 110 ±3 °F.
d. Condenser inlet.	Wet bulb - dry-coil condition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

TABLE XIII. Data required for run-in performance curves.

a. Dry-bulb temperature of air entering the evaporator, °F.
b. Wet-bulb temperature of air entering the evaporator, °F.
c. Dry-bulb temperature of air leaving the evaporator, °F.
d. Dry-bulb temperature of air entering the condenser, °F.
e. Discharge pressure, psig.
f. Suction pressure, psig.
g. Power consumption of the air conditioner, W.

4.5.2.14.2 Run-in performance curves. Not less than 25 units shall be used to establish acceptance curves (see 6.4). When less than 25 units are produced, all units will be used to establish acceptance curves. Data obtained in accordance with 4.5.2.14.1 shall be used by the contractor to select one high and one low unit that then shall be capacity-tested at condition "A" of 3.9.1. If any high- or low-limit air conditioner fails to meet minimum requirements of 3.9.1, it shall be replaced by the next low or high unit and that unit tested. Operate the acceptable high and low air conditioner continuously in the cooling mode for not less than 1 hour under each of the conditions specified in table XII. After stabilization, four sets of data specified in table XIII shall be recorded and averaged for each of the conditions in table XIV during the last 30 minutes (every 10 minutes). Establish high and low limits of acceptance for production units by plotting the following curves from data specified in table XIII:

a. Average evaporator-entering dry-bulb temperature versus average temperature differential between evaporator inlet and outlet air temperatures for each of the condenser-entering temperatures recorded.
b. Average evaporator-entering dry-bulb temperature versus power input for each of the condenser-entering temperatures.
c. Average evaporator-entering dry-bulb temperature versus suction pressure for each of the condenser-entering temperatures.
d. Average evaporator-entering dry-bulb temperature versus discharge pressure for each of the condenser-entering temperatures.

TABLE XIV. Temperature conditions for run-in performance curves.

	Temperature			
Location	Condition I	Condition II	Condition III	Condition IV
Evaporator return	87 °F ±0.5	93 °F ±0.5	87 °F ±0.5	93 °F ±0.5
Condenser inlet	107 °F ±0.5	113 °F ±0.5	113 °F ±0.5	107 °F ±0.5
Evaporator return and condenser inlet: Dry coil for all conditions.				

4.5.2.14.3 Failure criteria. One or more of the following shall constitute failure of this test:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

a. Temperature differential between the average air dry-bulb temperature entering the evaporator return and the average dry-bulb temperature leaving the evaporator before and after the 48-hour test phase in 4.5.2.14.1 not within the curves developed as specified in 4.5.2.14.2.
b. Average condensing pressure recorded before and after the 48-hour test phase in 4.5.2.14.1 not within the curves developed as specified in 4.5.2.14.2.
c. Average suction pressure recorded before and after the 48-hour test phase in 4.5.2.14.1 not within the curves developed as specified in 4.5.2.14.2.
d. Average power consumption on cooling recorded before and after the 48-hour test phase in 4.5.2.14.1 not within the curves developed as specified in 4.5.2.14.2.
e. Total power consumption (W) on heating (high-heat) in excess of that specified in 3.9.5.
f. Failure of any component or fracture or breakdown of any material.
g. Evidence of any detectable refrigerant leak.
h. Need for any servicing of the air conditioner during operation.

4.5.2.15 Reliability. Using the upper and lower tests MTBFs specified in 3.13, the air conditioner shall be tested as specified herein with "accept" and "reject" criteria in accordance with MIL-STD-781, test plan VIIID. Each air conditioner tested must accumulate at least one-third of the total test hours.

4.5.2.15.1 Test facility. Install the air conditioners in a test chamber consisting of two rooms. One room shall function as an indoor room and shall maintain the evaporator return-air temperature within ±5 °F, as specified on figure 5. The second room shall function as an outdoor room and shall maintain the condenser return air temperature as specified on figure 5 within ±5 °F. These temperatures shall be maintained on a 24-hour cyclic basis throughout the reliability test, as shown on figure 6.

4.5.2.15.2 Operation. All air conditioner controls (rotary switch, toggle switch, and thermostat) shall be operated manually or automatically. If automatic means are used, they shall not interrupt the control circuit but shall physically operate the air conditioner control components. The controls shall be positioned as shown in table XV.

TABLE XV. Control positions for reliability test.

Mode	Time	Control	Control Position
Heating	0030 to 0130	Mode switch	High heat
Heating	0030 to 0130	Thermostat	Warmest
Heating	0100 to 0130	Thermostat	Coolest
Bypass	0130 to 0400	Mode switch	Cool
Bypass	0130 to 0400	Thermostat	Warmest
Cooling	0400 to 2100	Mode switch	Cool
Cooling	0400 to 2100	Thermostat	Coolest
Bypass	2100 to 2330	Mode switch	Cool
Bypass	2100 to 2330	Thermostat	Warmest
Off	2330 to 0030	Mode switch	Off

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.5.2.15.3 Data. The following shall be monitored and recorded throughout the test as required to determine that normal air conditioner operation is being maintained and to identify problems in event of failure or malfunction:

a. Voltage.
b. Current.
c. Frequency.
d. Refrigerant pressures.

    (1) Suction.
    (2) Discharge.

e. Refrigerant temperatures.

    (1) Compressor discharge.
    (2) Liquid line.
    (3) Suction line (at compressor).

f. Air temperatures.

    (1) Evaporator return (dry bulb).
    (2) Evaporator discharge (dry bulb).
    (3) Condenser return (dry bulb).

g. Abnormal operation (visual and audio observation).
h. Sight-glass observation (clear or bubbles).
i. Identification of unit.
j. Initials of technician.
k. All failures.
l. Time of day of failure.
m. Action taken if failure occurs.

4.5.2.15.4 Failure criteria. A failure is defined as any malfunction or deficiency that cannot be deferred until the next scheduled maintenance or cannot be repaired at the organizational level within 30 minutes. Malfunctions will not be considered "failures" for reliability purposes if detected and corrected during initial inspection or if they may be deferred to final inspection (except for those that cause test termination). Malfunctions that result from operational or maintenance errors will not be counted. A malfunction shall be considered to have occurred if the following happens:

a. Temperature difference across the evaporator coil is lower than specified in table XVI when the air conditioner is in the cooling mode.
b. Current draw is greater than specified in table XVI when the air conditioner is in the cooling mode.
c. Current draw is greater than specified in table XVI when the air conditioner is in the heating mode with the thermostat in the extreme clockwise position.
d. Current draw does not decrease when the air conditioner is in the heating mode and the thermostat is changed from the extreme heating position to the extreme cooling position.
e. Temperature at the evaporator discharge does not increase when the air conditioner is turned to the heating mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

f. The sight-glass is not clear when the air conditioner is in the cooling mode.
g. The discharge pressure is greater than specified in table VI (condensing pressure column under 3.9.1) when the air conditioner is in the cooling mode.
h. The suction pressure is lower than specified in table VI (suction pressure column under 3.9.1) when the air conditioner is in the cooling mode.
i. The suction pressure is lower than 35 psig when the air conditioner is in the bypass mode.
j. The evaporator discharge temperature does not increase when the air conditioner is changed from the cooling mode to the bypass mode.

TABLE XVI. Minimum temperature drop and maximum current draw (see note).

Size	Class	Temperature difference °F (minimum)	Current draw Amps (max)
Type I			
A	1 2	13.5 12.6	21.3 14.5
B	1 2 3 4	17.1 17.1 17.1 17.1	33.0 15.6 11.3 18.2
C	1 2	20.0 20.0	15.5 26.0
D	1 2	18.2 16.7	32.2 50.9
E	1 2	19.0 17.9	55.2 82.0
Type II			
A	1 2 4	18.0 18.0 18.0	33.0 15.6 21.6
B	1 2 3	20.0 20.0 20.0	29.8 22.6 32.9
C	1 2	20.0 19.5	45.2 33.6
D	1 2	22.0 22.0	45.8 74.6

Note: For all 3-phase air conditioners, amperage reading shall be the average of the three phases.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

k. The sight-glass does not empty at the time the air conditioner (type I units only) is changed from the cooling mode to the bypass mode.
l. Any abnormal operation occurs as determined by looking at or listening to the unit.

4.5.2.16 Transportability. The air conditioners shall be mounted and transported as specified in 3.16. Nonconformance to 3.16 shall constitute failure of this test.

4.5.2.17 Maintainability. The maintenance ratio shall be computed as specified in 3.17 for all maintenance actions required during or as a result of first article testing. Nonconformance to 3.17 shall constitute failure of this test.

4.5.2.18 Safety. The air conditioner shall be evaluated throughout the testing as specified in 4.5.2. Nonconformance to 3.18 shall constitute failure of this test.

4.5.2.19 Blast damper. The blast dampers shall be tested on an operating air conditioner with the evaporator side connected to a space that is to remain essentially at normal atmospheric pressure. The three sides and the top of the air conditioner shall then be subjected to three series of blast pressures. Each series shall consist of a sufficient quantity of blast pressures to ascertain that the dampers remain open at 2.5 inches water gauge (wg), close at 3.6 wg, and can withstand a minimum of 3.4 psig.

4.5.2.19.1 Basis of failure. One or more of the following shall constitute failure of the blast damper test:

a. Failure to remain open at 2.5 inches wg.
b. Failure to close at 3.6 inches wg.
c. Evidence of permanent deformation, rupture, or leakage.

4.5.2.20 Noise limits. Measure the noise level of the air conditioner in the "cool" mode in accordance with MIL-STD-1474. General test requirements (MIL-STD-1474, 5.1.2.1) shall be:

a. Measurement location: see figure 7
b. Record:
   (1) octave band level and dB(A) levels at all locations every 10 minutes for 30 minutes (4 sets)
   (2) background level, immediately before start and after completion
c. Noise contour: MIL-STD-1474, 5.1.2.1.4
d. Equipment openings:
   Evaporator and condenser grilles - open (no ductwork).
   Ventilation damper - closed.
   CB opening, if any - closed.
e. Duty cycle: continuous operation in cooling (compressor operating, fan high speed if applicable).
f. Test site: see figure 7.
g. Data recording: conforming to MIL-STD-1474, 5.5.

Nonconformance to 3.20 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

4.6 Inspection of packaging.

4.6.1 First article pack inspection. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects.

124.	Louvers not closed and canvas covers, if supplied, not snapped in place as specified for level A, B, or commercial.
125.	Technical publications not preserved and placed with the air conditioner as specified for level A or B.
126.	Accessories not cushioned, placed in bag or box and positioned as specified for level A or B.
127.	Sound attenuators, when included, not prepared as specified for level A or B.
128.	Air conditioner not preserved as specified for level A or B.
129.	Air conditioner shipping container not as specified for level A or B.
130.	Cushioning not of the type, size, or positioned as specified for level A or B.
131.	Sound attenuator, when included, not packed inside shipping container with air conditioner, as specified for level A or B.
132.	Strapping not as specified for level A or B.
133.	Clearance between air conditioner and inner faces of box not as specified for level A or B.
134.	Air conditioners requiring bolting to the base of the box not bolted as specified for level A or B.
135.	Wood spacers, when required, not padded as required for level A or B.
136.	Air conditioners not preserved, packed in accordance with the reference document for commercial.
137.	Palletization, when appropriate, not as specified for all degrees of packaging.
138.	Marking missing, illegible, incorrect, or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.8). The first article pack may be accomplished utilizing either the first article model air conditioner or a production air conditioner. If the first article model air conditioner is utilized, any preservation, and packing

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model air conditioner and production air conditioners.

5.2 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 General.

5.2.1.1.1 Canvas covers and louvers. Prior to applying the method II barrier material as required in 5.2.1.2 or 5.2.1.3, around the air conditioner, all louvers shall be placed in their closed position. Louver control chains, if so equipped, shall be secured in place. Canvas covers, if supplied, shall be snapped in place.

5.2.1.1.2 Technical publications. Technical publications shall be preserved by method IC-1 or IC-3 and placed with the air conditioner within the method II barrier specified in 5.2.1.2 or 5.2.1.3.

5.2.1.1.3 Accessories.

a. Block-off assemblies and cable connectors shall be placed together in a sack conforming to A-A-1588, which shall be secured shut with staples. For the type II air conditioners, parts of the vibration mounts shall be placed with the above accessories. Cable connectors shall be wrapped with cushioning material conforming to A-A-1898 prior to placement in the sack with other accessories. The consolidated accessories shall be secured with tape to the air conditioner in a location which will not interfere with any cushioning or blocking; however, on type I, size C air conditioners, the accessories shall be placed behind the junction box in the condenser compartment.
b. Control adapter assemblies, power adapters, blast damper assemblies, and fresh air covers, if provided, shall be individually packed in boxes conforming to PPP-B-636, type CF, class domestic, variety SW, weight appropriate for the weight of contents. Prior to placement in the boxes, each accessory shall be wrapped with cushioning material conforming to A-A-1898. Special air conditioning covers, if provided, shall be placed in a polyethylene bag having a film thickness of not less than 2.5 mils. The bag shall be secured shut with staples or by heat sealing.

5.2.1.1.4 Sound attenuators. When sound attenuators are included, they shall be placed in boxes conforming to PPP-B-636, type CF, class weather resistant, variety SW, grade appropriate for the weight of the attenuator.

5.2.1.2 Type I, sizes A and B; and type II, size D air conditioners. Air conditioners shall be preserved in accordance with MIL-P-116, submethod IIe. The housing of the air conditioner shall serve as the inner container of the preservation method.

5.2.1.3 Type I, sizes C, D, and E; and type II, sizes A, B, and C. Air conditioners shall be preserved in accordance with MIL-P-116, submethod IIa. Air conditioners shall be secured to the bottom of boxes specified in 5.3.1.3 with four bolts which pass through the barrier material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

5.2.2 Level B.

5.2.2.1 General. All louvers shall be closed, and canvas covers, when supplied, shall be snapped in place. Technical publications and accessory items (block-off assemblies, cable connectors, vibration mounts, etc.) shall be preserved and secured to the air conditioner as specified for level A. Sound attenuators shall be packed as specified in 5.2.1.1.4 except that boxes shall be domestic class and of a grade appropriate with the weight and size of the attenuator.

5.2.2.2 Air conditioners. Air conditioners shall be covered with a polyethylene film shroud of not less than 3 mils of thickness. The shroud shall be secured to the air conditioner with tape.

5.2.3 Commercial. Each complete air conditioner with louvers closed and canvas covers, if supplied, snapped in place shall be preserved in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Type I, sizes A and B. Air conditioners, preserved as specified in 5.2, shall be packed in a box conforming to PPP-B-601, overseas type, style I. The box shall be of a size into which the air conditioner with sound attenuator, and other included accessories, and any necessary blocking will fit snugly. Fiberboard corner pads of the accordian-folded or multiple layer type conforming to MIL-STD-1186 shall be used on all top and bottom edges of the air conditioners. Fiberboard corner pads shall not be less than 1 inch thick and 4 inches wide. When a wound attenuator is included, it shall be placed alongside the air conditioner. Blocking or cushioning conforming to MIL-STD-1186 shall be used to prevent movement of the packed attenuator. The box shall be closed and reinforced in accordance with the box specification. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable.

5.3.1.2 Type II, size D. Air conditioners with all included accessories, preserved as specified in 5.2, shall be packed in a container consisting of a plywood base and a wood-cleated plywood cover as shown on figure 8. The air conditioner shall be cushioned with expanded polyethylene as shown on figure 9. The cushioning material shall conform to PPP-C-1752 with nominal density of 2.2 pounds per cubic foot. Strapping shall be as specified in 5.3.1.1.

5.3.1.3 Type I, sizes C, D, and E; type II, sizes A, B, and C. Air conditioners with all included accessories preserved as specified in 5.2 shall be packed in a box conforming to PPP-B-601, overseas type. The box size shall be such that not less than 1 inch clearance shall exist between faces of the air conditioner and the inner walls of the box. The air conditioner shall be secured to the bottom of the box with four bolts, utilizing the boltholes provided in the bottom of the air conditioner. The box shall be modified with skids in accordance with the box specification except that the skids shall be located so that the bolts securing the air conditioner pass through the skids. Boltheads shall be countersunk into the bottom of the skids. Strapping shall be as specified in 5.3.1.1. When a sound attenuator is included with a type I air conditioner, it

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

shall be placed alongside the air conditioner. Movement of the attenuator shall be prevented by the use of blocking or cushioning meeting the requirements of MIL-STD-1186. On type I sizes D and E air conditioner, wood or weather resistant fiberboard spacers, the full width and depth of the unit, shall be placed on all sides of the unit at the top edges to provide support between the unit and box walls. All wood spacers shall be padded with water resistant cushioning material conforming to A-A-1898 to prevent abrasion of the barrier material.

5.3.2 Level B.

5.3.2.1 Type I, sizes A and B, C, D and E, and type II, sizes A, B, and C. Air conditioners and all accessories preserved as specified in 5.2 shall be packed as specified in 5.3.1.1 and 5.3.1.3 except the boxes shall be domestic type and required strapping shall be coated finish or the air conditioners shall be packed in a container as shown in figure 10. Corner pads, and any required blocking or cushioning for the sound attenuator, when included, shall be as specified in 5.3.1.1. The box shall be closed in accordance with the appendix to the box specification.

5.3.2.2 Type II, size D. Air conditioner and all accessories preserved as specified in 5.2 shall be packed and cushioned as specified in 5.3.1.2 and figures 8 and 9. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.3.3 Commercial. Air conditioners, preserved as specified in 5.2, shall be packed in containers in accordance with ASTM D 3951.

5.4 Palletization. When a multiple quantity of air conditioners is being shipped to the same destination, packed units for all levels shall be palletized. Only those air conditioners whose size will permit two, three, or four units on a 40 by 48 inch pallet will be palletized. Pallets for levels A and B shall conform to NN-P-71, 40 by 48 inches, type optional. Commercial packed units may utilize a commercial type 40 by 48 inch pallet suitable for the weight involved. Air conditioners shall be secured to the pallet with strapping conforming to ASTM D 3953, type 1 or 2, size as applicable. For level A palletized loads strapping shall be zinc-coated. Strapping shall be arranged so that each air conditioner on the pallet is restrained in all directions..

5.5 Marking.

5.5.1 Military. In addition to any special marking as specified in the purchase order or the contract (see 5.5.3), marking shall be in accordance with MIL-STD-129.

5.5.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951, and the special marking specified in 5.5.3. In addition, weight and cube data shall be marked on each shipping container.

5.5.3 Special marking (see 6.2). Each shipping container shall be marked to indicate that the air conditioners shall remain in the operating position during shipment and storage. Special markings shall consist of arrows and the words "THIS END UP". Special marking shall be placed within the upper third of each side of the shipping container in black letters not less than 3 inches high.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The air conditioners covered by this specification are intended for use in cooling, dehumidifying, heating, filtering, and circulating air for portable shelters or vans in accordance with electronic and personnel requirements. The air conditioner may be installed indoors or directly exposed to the weather.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type, size, class and part or identifying number of air conditioner required (see 1.2 and 6.6).
d. When a block-off assembly is required for type I (see 3.1).
e. When external sound attenuators are required for type I (see 3.1).
f. When a control adapter assembly and power adapter are required (see 3.1).
g. When a nuclear hardening assembly is required for type II, size B (see 3.1).
h. When a blast damper assembly, fresh-air cover, and special air conditioner cover for use with blast damper assembly are required (see 3.1).
i. Time frame required for submission of first article and number of units required (see 3.2 and 6.3).
j. Color required when other than camouflage green (see 3.15).
k. Time frame required for submission of the first article pack (see 5.1).
l. Degree of preservation and packing required (see 5.2 and 5.3).
m. Any special marking required (see 5.5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of five units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

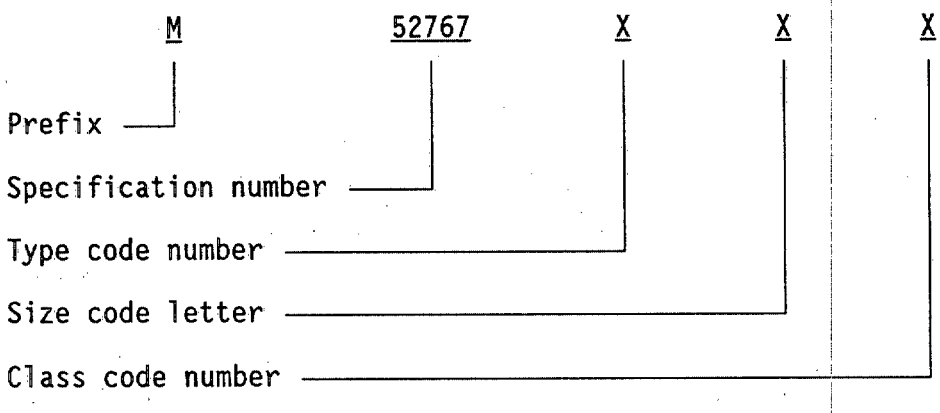
6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each air conditioner. In addition, the contracting officer should require that copies of the performance curves specified in the run-in test be furnished to the engineering agency within 30 days after development.

6.5 Accessory uses.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

a. A control adapter assembly (see 3.1) shall be required when type I, size D air conditioners are furnished for use with control panels produced as part of air conditioner model number A/E32C-24 and A/E32C-25.
b. A power adapter (see 3.1) shall be required when type I, size D air conditioners are furnished for use with power cables previously fabricated to provide power to air conditioner model numbers A/E32C-24 and A/E32C-25.
c. Blast damper assemblies, fresh-air covers, and special air conditioner covers shall be required when type I, size C, class 1 air conditioners are produced for Pershing II systems.
d. Nuclear hardened assembly shall be required when type II, size B, class 2 air conditioners are produced for systems with nuclear hardening requirements.

6.6 Part or identifying number (PIN). The PIN to be used for air conditioners acquired to this specification are created as follows (see 1.2):



6.7 Welders and welding operators. The contracting officer should consider requalifying welders or welding operators in the event of evidence of poor welds (see 3.21.4.1).

6.8 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the obligation of the contractor to preserve, pack and mark the air conditioners in accordance with this specification.

6.9 Subject term (key word) listing.

Circulating air
Cooling air
Dehumidifying air
Environmental control unit
Heating air
Shelters, portable
Vans, portable

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

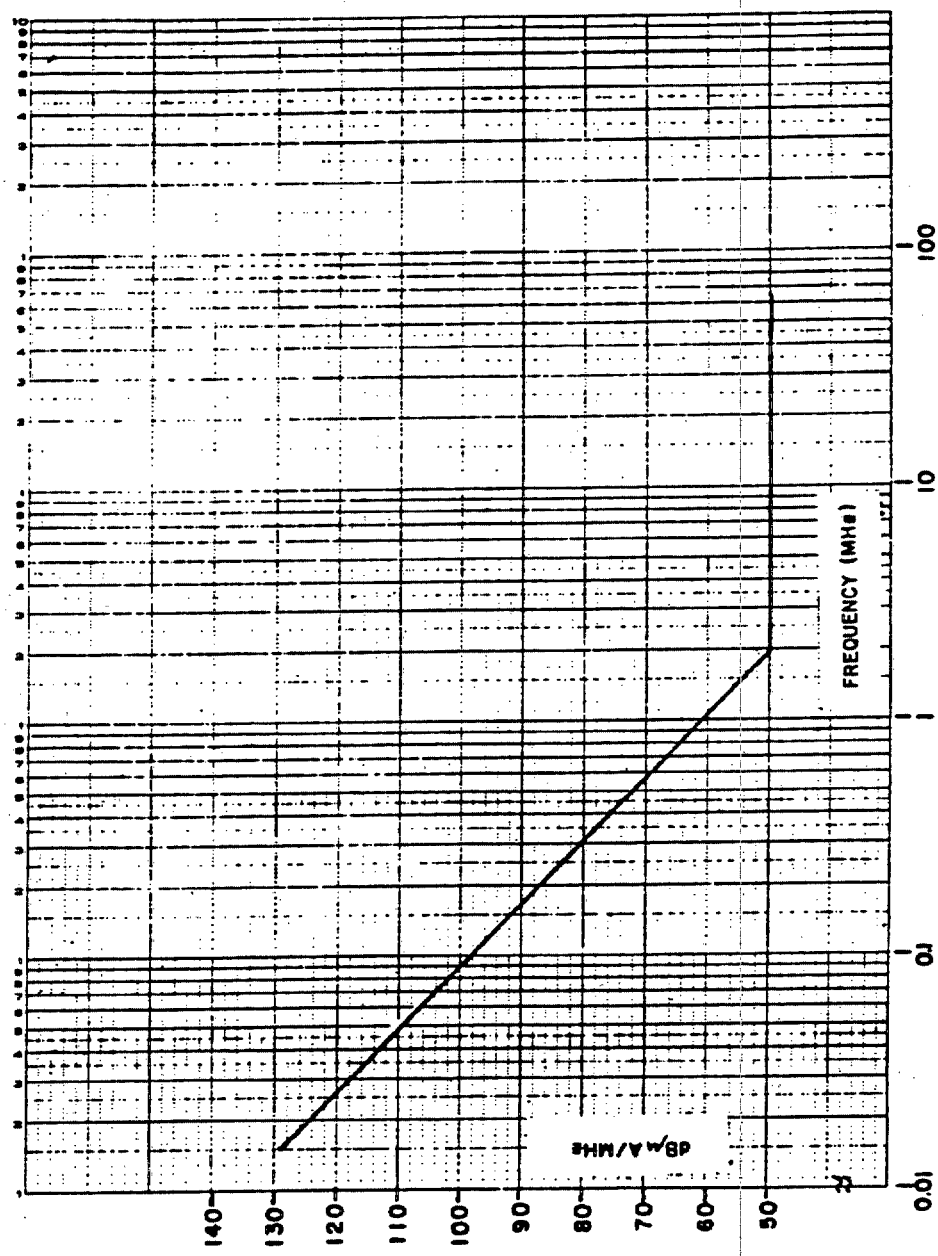


FIGURE 1. Limit for CE03 broadband emissions DC and interconnecting leads and AC leads.

X-3747A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

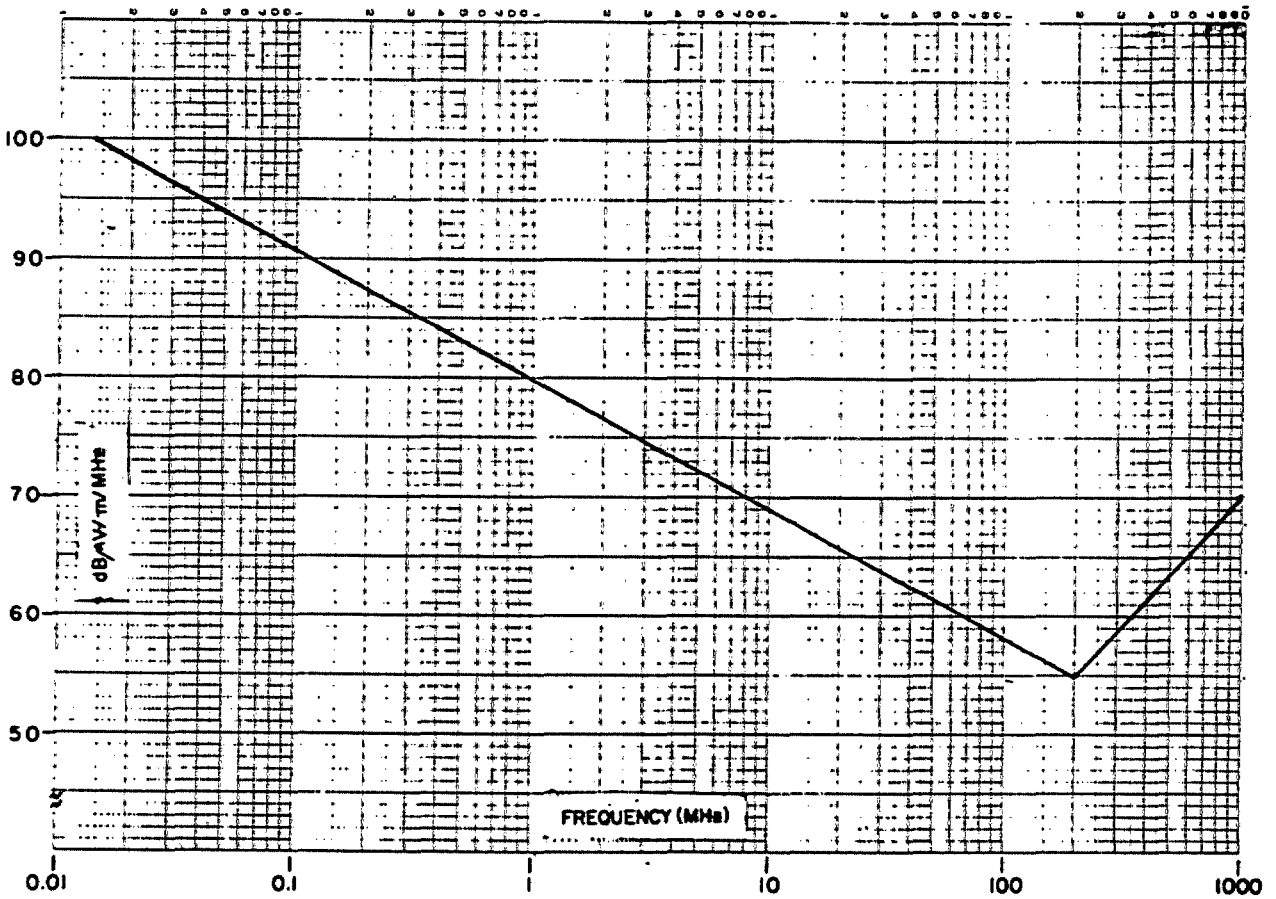


FIGURE 2. Limit for REO2 broadband emissions.

X-3748

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

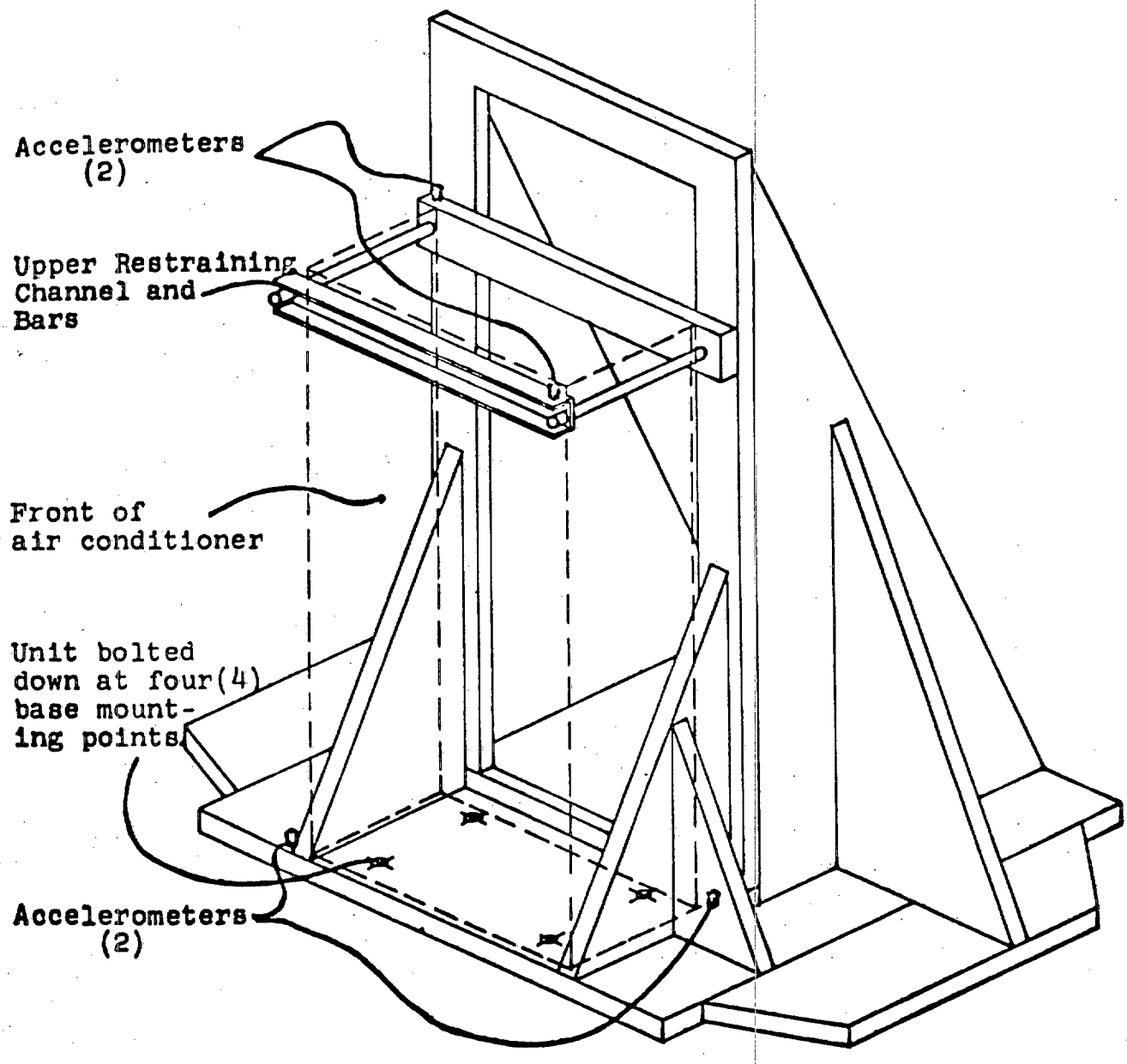
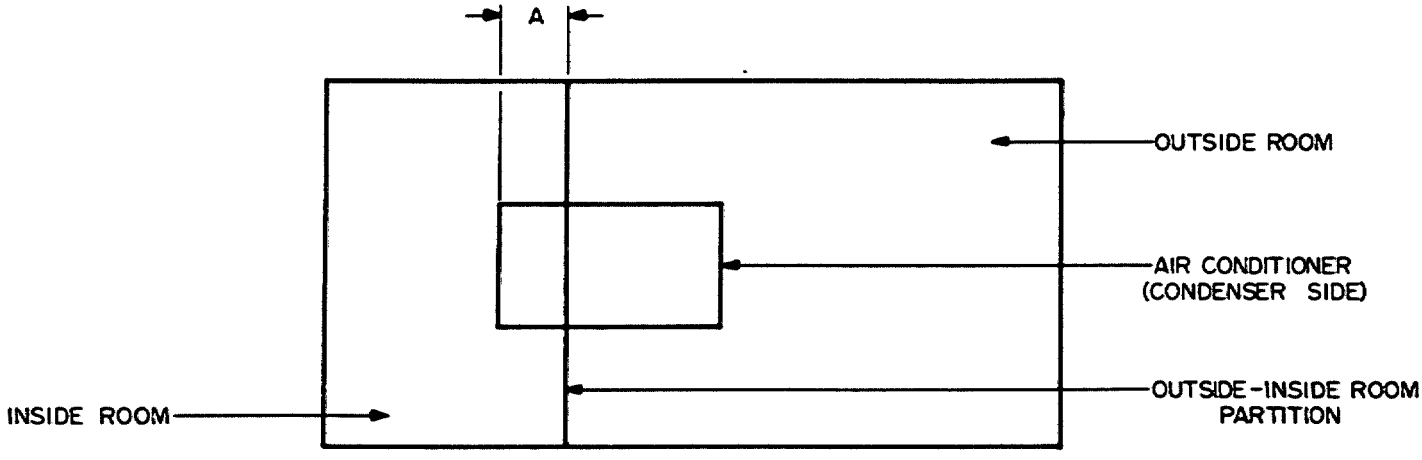


FIGURE 3. VIBRATION TEST FIXTURE X-2170A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

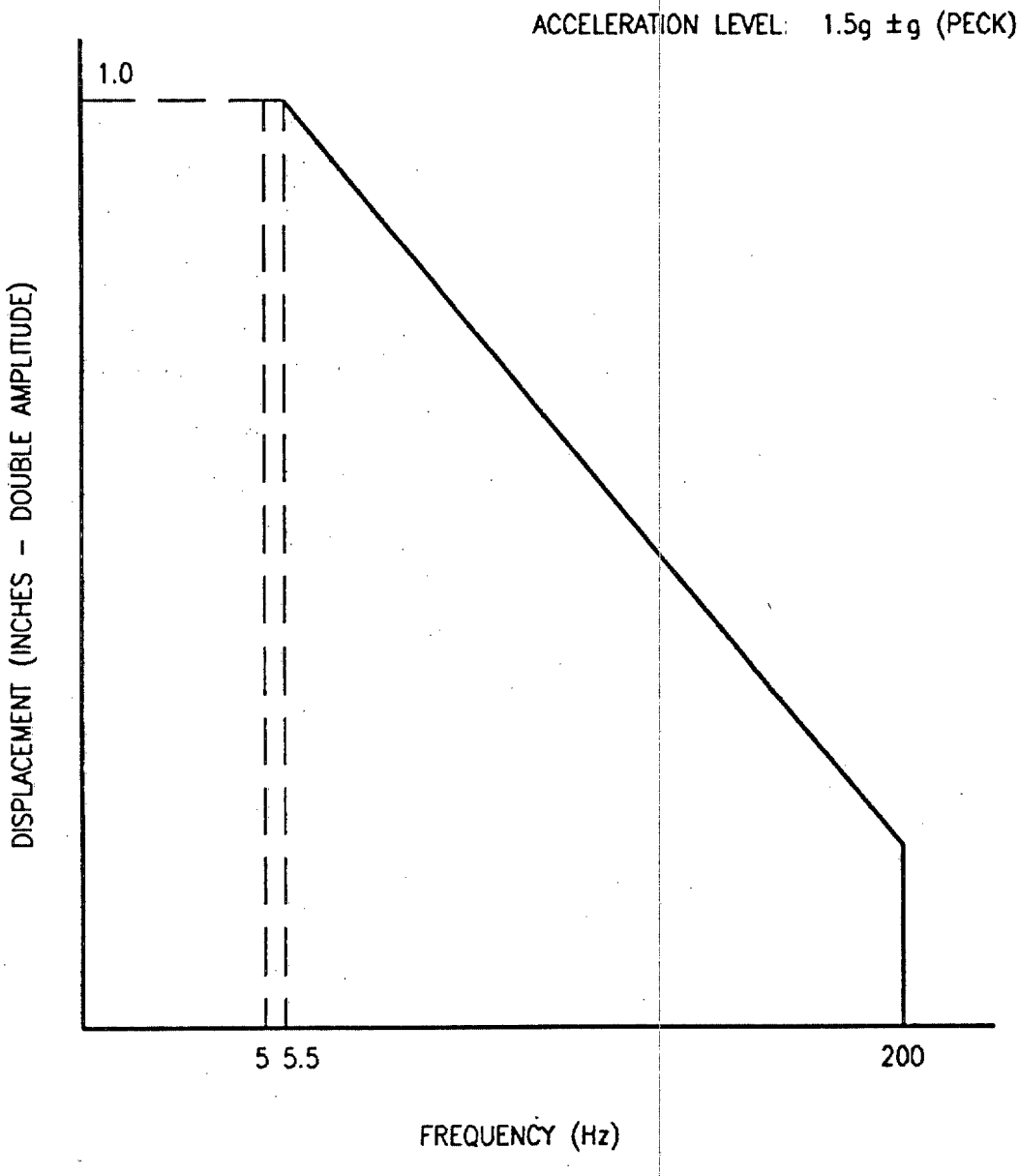


AIR CONDITIONER		"A" DIMENSION
TYPE	SIZE	INCHES
II	B	11.50 ± 0.5
II	C	14.75 ± 0.5

FIGURE 4. Type II air conditioner locations for capacity test.

X-2440B

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



Note: Curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected or measured and sweep time shall be 18 min.

FIGURE 5. Vibration test curve for equipment installed in ground vehicles.

X-5034

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

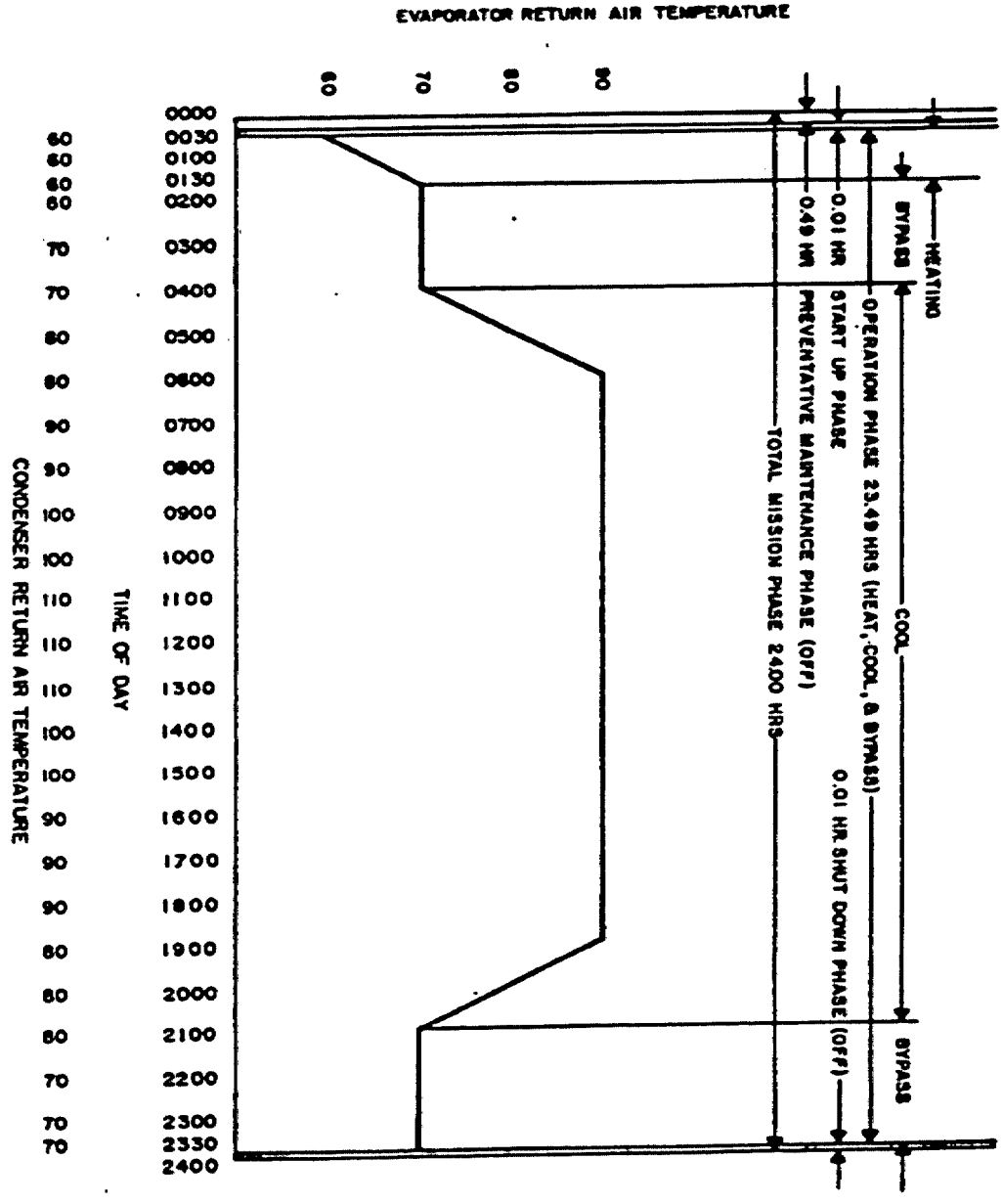
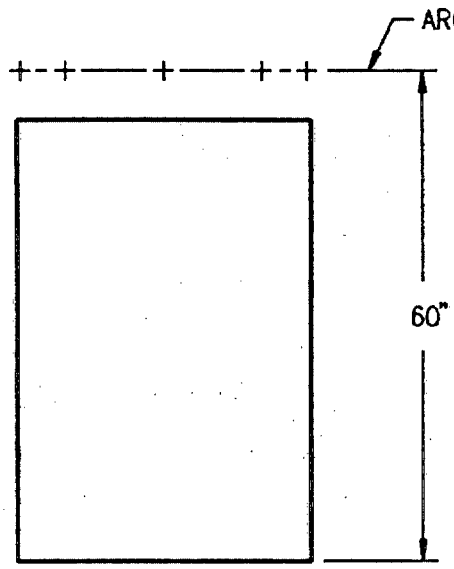


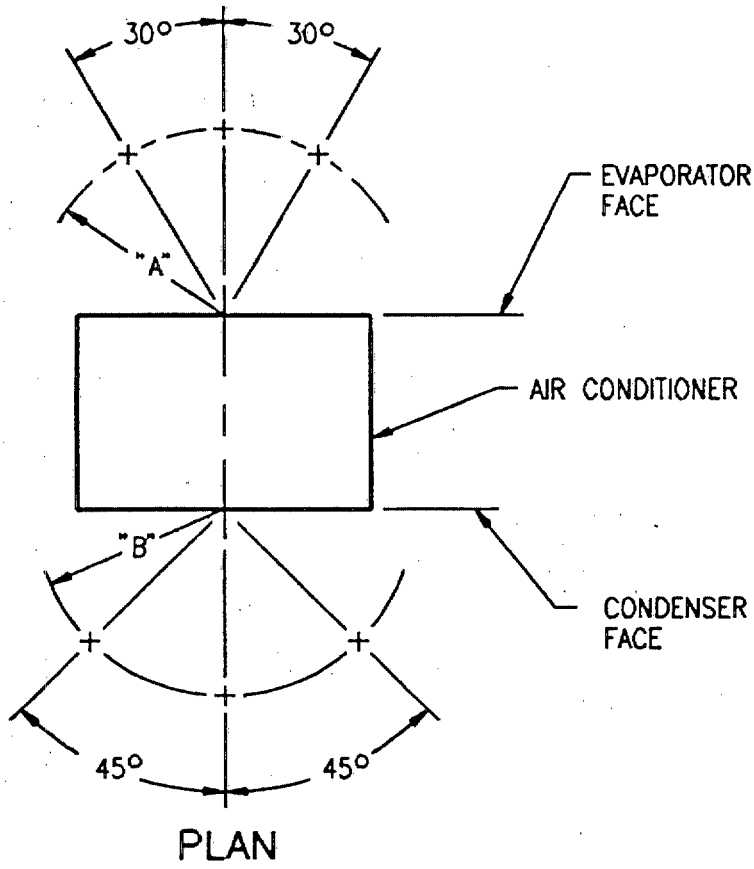
FIGURE 6. STANDARDIZED TEST PROFILE.

X-2441

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



ELEVATION



PLAN

AIR COND		TEST	DIM	
TYPE	SIZE	SITE	"A"	"B"
I	A	*	48	48
I	B	*	48	48
I	C	**	36	60
I	D	*	48	48
I	E	*	48	48
II	A	**	36	60
II	B	**	36	60
II	C	**	36	60
II	D	*	48	48

* — CONDITIONED AREA OF PROPERLY ENGINEERED INSTALLATION

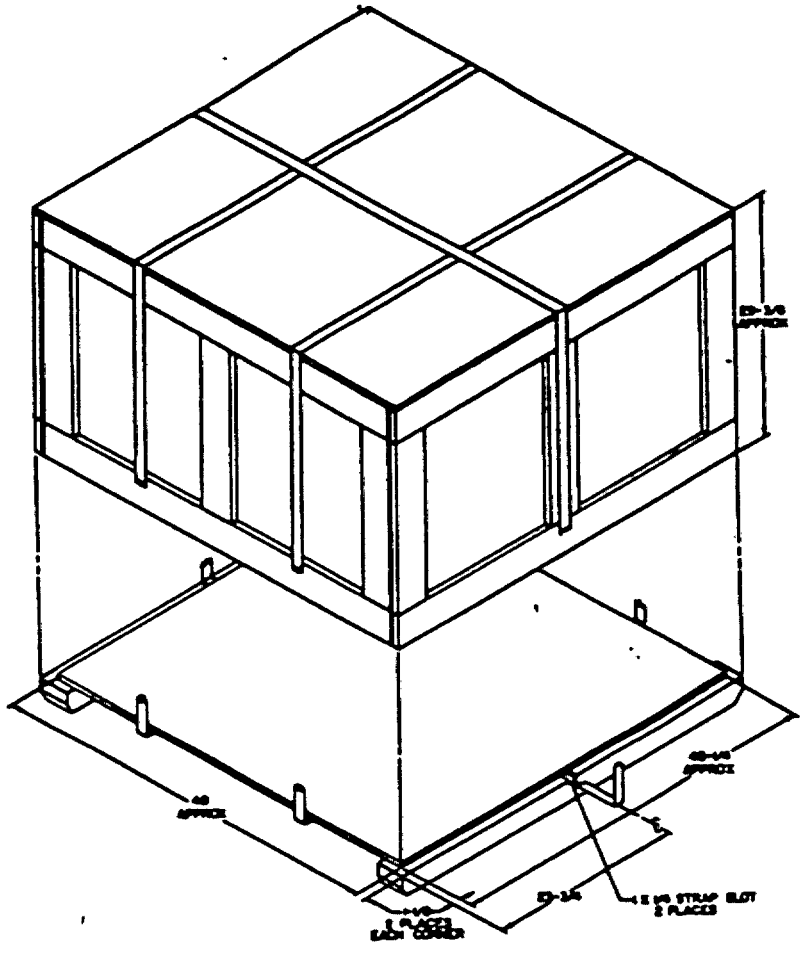
** — ANECHOIC CHAMBER

\+ — MICROPHONE LOCATION

FIGURE 7. Noise Contour.

X-5035

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



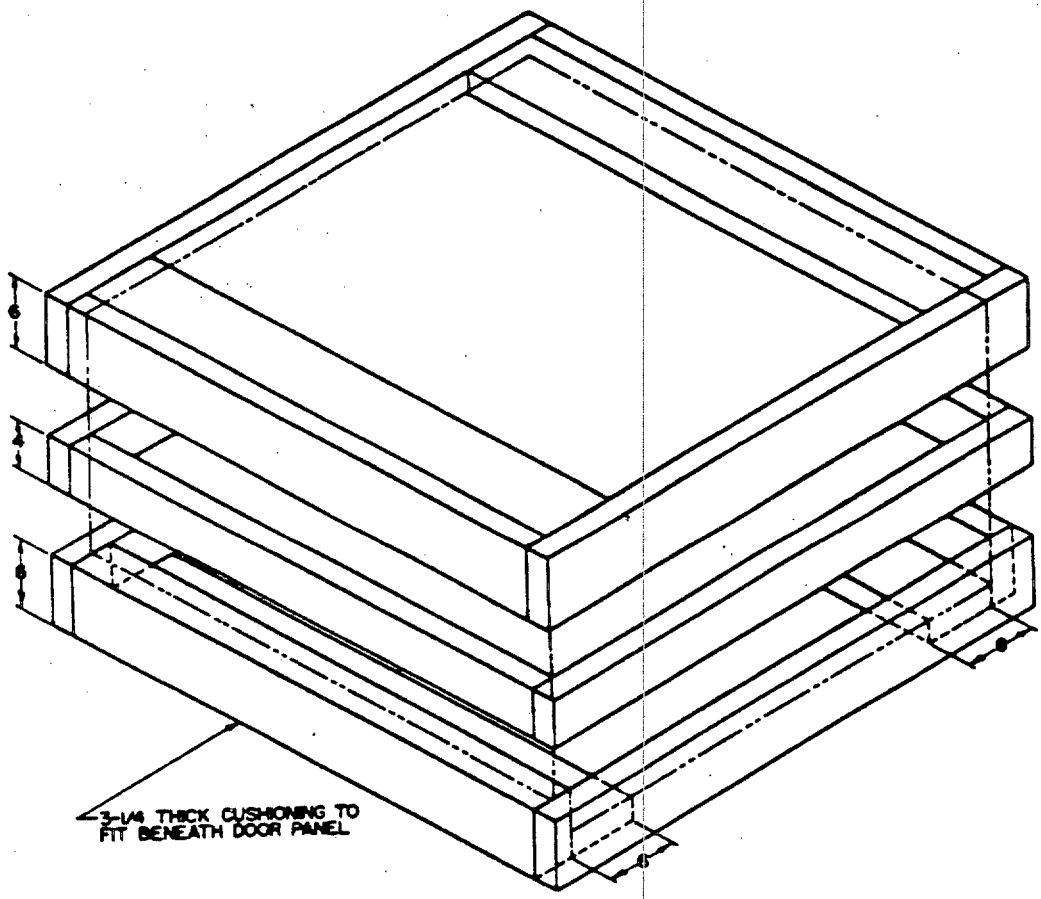
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. PLYWOOD SHALL CONFORM TO PS 1, GRADE C-D INTERIOR WITH EXTERIOR GLUE; OR TO ANSI/HPMA HP 1983, GRADE 3-4, TYPE I. IT SHALL BE OF THE FOLLOWING MINIMUM THICKNESSES: BASE-1/2; COVER-3/8.
3. NAILS SECURING PLYWOOD FLOOR TO SKIDS SHALL CONFORM TO FF-N-105, TYPE II, STYLE 18, 2-1/2 INCHES LONG, SPACED FOUR INCHES APART AND STAGGERED.
4. CLEATS SHALL CONFORM TO THE REQUIREMENTS OF PPP-B-601.
5. TOP SHALL BE A SINGLE PIECE OF PLYWOOD, WITHOUT CLEATS.
6. THE EXACT SIZE OF THE BASE AND COVER SHALL BE DETERMINED BY ACHIEVING A SNUG FIT OVER THE CUSHIONING AROUND THE AIR CONDITIONER.
7. ASSEMBLY NAILING OF COVER SHALL BE IN ACCORDANCE WITH PPP-B-601.

X-2324B

FIGURE 8.. Shipping container levels A and B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.



NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. CUSHIONING MATERIAL SHALL BE TWO INCHES THICK WITH THE EXCEPTION SHOWN.
3. SECURE CUSHIONING TO UNIT WITH TAPE TO PREVENT MOVEMENT WHILE INSTALLING BOX COVER.

X-2325

FIGURE 9. Cushioning design levels A and B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

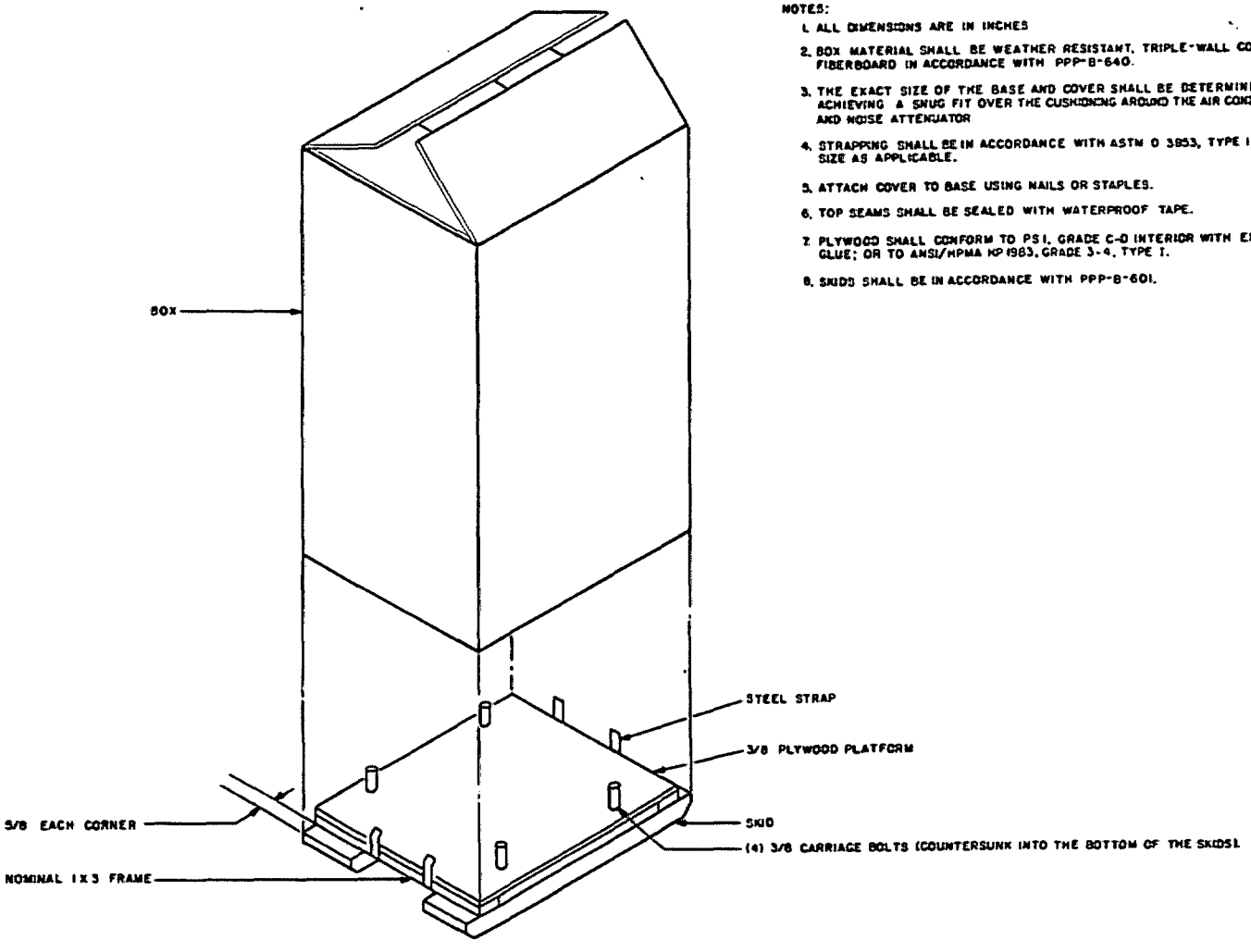


FIGURE 10. Shipping container Level B.

X-3597B

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-0052767E(ME)	2. DOCUMENT DATE (YYMMDD) 940510

3. DOCUMENT TITLE
AIR CONDITIONERS: VERTICAL AND HORIZONTAL, COMPACT

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) DSN 654-3468
c. ADDRESS (Include Zip Code) US ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:10Z
Check the source to verify that this is the current version before use.